UX CHALLENGE · NEW AI ASSISTANT FOR INFOTAINMENT

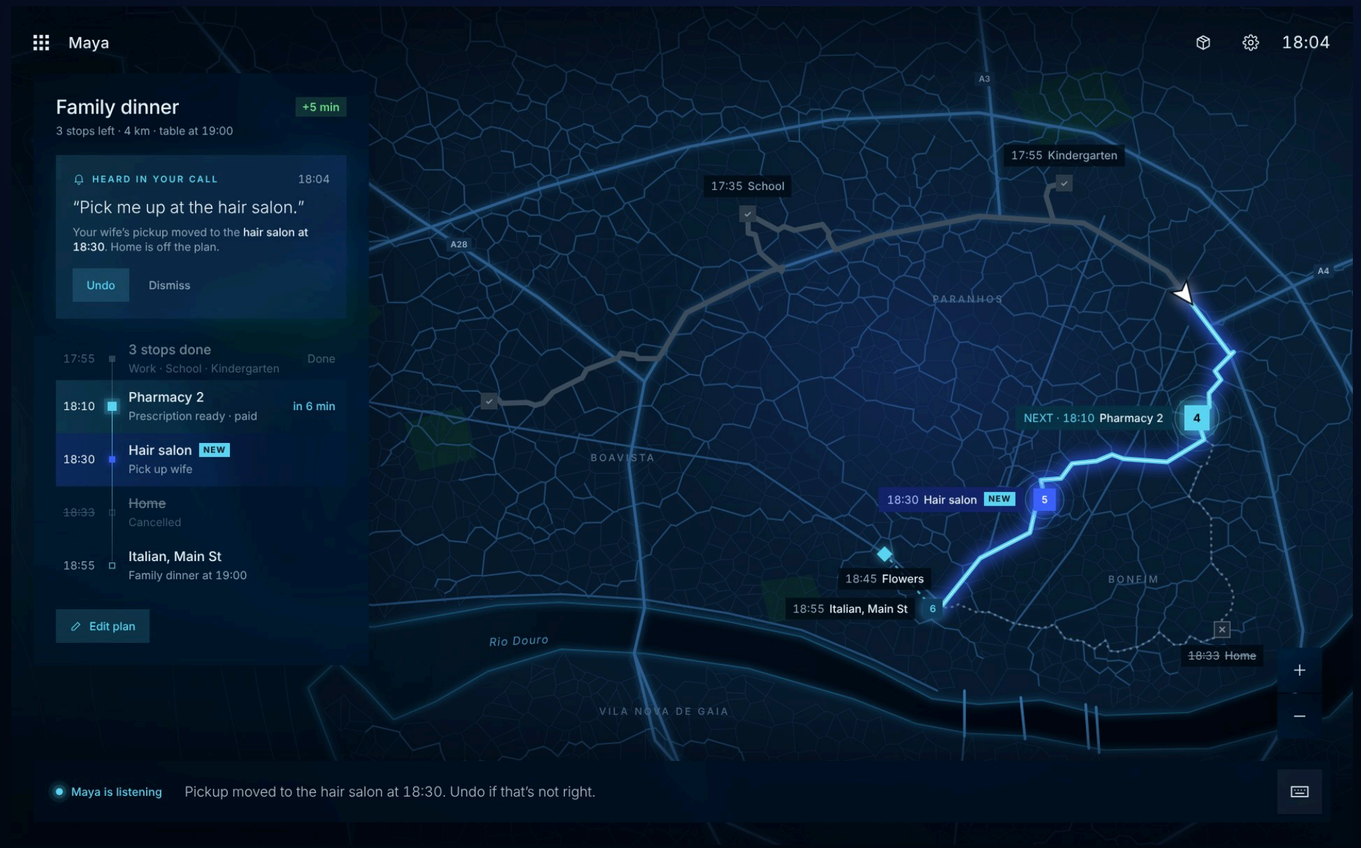
Maya

L5 trip concierge

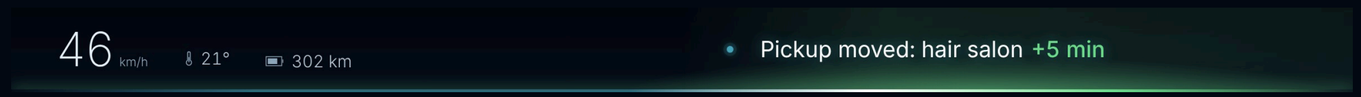
A self-driving car frees your hands, not your agenda. Navigation handles routes, not commitments, so passengers still sequence the stops, phone businesses and replan whenever something changes.

With Maya, you share your needs and goals and she plans the stops, confirms with the shops, pays from her wallet and adapts when conditions change. Everything is shown on your central display and Panoramic Vision.

When the car is free, she becomes the courier that runs your errands.



CENTRAL DISPLAY



PANORAMIC VISION

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06 **Design system** — p. 24–26
Foundations, components and more

07 **Rules and validation** — p. 27–29
Edge cases · validation and metrics · trade-offs and what's next

WORKING ALONE

A solo exercise with no user research. The assumptions are written down, ranked by risk, and paired with the tests that would check them.

Each artifact answered one question before the next one started.

01	**Concept foundations**	To commit to one bet and rank what could break it, so the riskiest parts (ambient intent, AI callers, an empty car) get designed and tested first.
02	**The scenario**	An assistant is mostly conversation. Writing every line first decided when Maya speaks, when she stays silent and what the screen must show instead.
03	**Journey map**	To see the whole evening at once: Michael's goals and feelings, what Maya does backstage and where the risks sit.
04	**Task flow**	The logic behind every replan and the courier hand-off, one decision at a time.
05	**Screen specs and mood board**	To fix the screen inventory and IDs, decide what Panoramic Vision may show, and choose a direction: Deep ocean, because in a dark cabin light can carry meaning.
06	**Multimodal matrix**	To keep voice and touch equal: every voice action has a touch twin, every spoken line a caption.
07	**Edge cases and validation plan**	To design for the evening going wrong, and to say how each assumption would be tested and what success looks like.
08	**Central display wireframes**	To settle layout and hierarchy (plan on the left, map behind, Maya below) before any visual decisions.
09	**Hi-fi screens and components**	The whole story in final quality, built from one set of components so every state stays consistent.
10	**Clickable prototype**	To feel the timing of the story, and to run the remote tests in the validation plan.

THREE LINKS, ALL OPTIONAL

Plain web pages, no sign-in. This PDF stands on its own; the links add full-size screens, the working files and the prototype.

ENTRY PAGE

oliveiraanibal.github.io/maya/

This PDF and every working artifact, in the order they were made.

SCREENS

oliveiraanibal.github.io/maya/screens/

All hi-fi screens at full size, the Panoramic Vision states and the component inventory.

PROTOTYPE

oliveiraanibal.github.io/maya/prototype/

The whole evening by touch or voice, with edge cases 1 to 4 as what-if branches.

A self-driving car frees your hands, not your agenda.

Navigation handles routes, not commitments, so passengers still sequence the stops, phone businesses and replan whenever something changes. With Maya, you share your needs and goals and she plans the stops, confirms with the shops, pays from her wallet and adapts when conditions change. Everything is shown on your central display and Panoramic Vision. When the car is free, she becomes the courier that runs your errands.

INSIGHT

People think in needs, not addresses

“Family dinner tonight. Let’s get the kids and my wife, and I need that new prescription.” One sentence holds three pickups, a pharmacy to find and pay, a restaurant and a deadline.

VISION

“I share my needs and goals; everything else just happens, and I can see it happening.”

“With me in the car, Maya runs my trip. Without me, my car runs my errands.”

HYPOTHESIS

Time-poor people will hand trip logistics to an AI that **asks only what it cannot know**, acts on its own, follows the conversation and makes every action **visible and reversible**.

We are right when people arrive on time, rarely undo what Maya did, and trust her with more over time.

WHO

Michael, a father of two with a packed evening. Maya belongs to him, not to the car: she is in any car he signs in to, his own or a robotaxi.

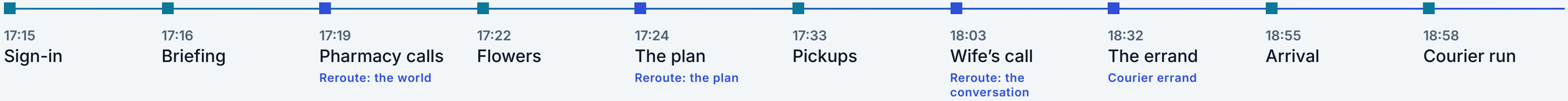
ASSUMED, NOT DESIGNED

Payments from Maya Wallet up to a limit he sets. Schools release the children to the car. The florist delivers to the restaurant.

ACCESSIBLE FROM THE START

Every task works with voice alone or touch alone. Everything Maya says is captioned, and colour never carries meaning on its own.

THE HERO SCENARIO: THE FAMILY DINNER RUN, ONE EVENING IN TEN SCENES



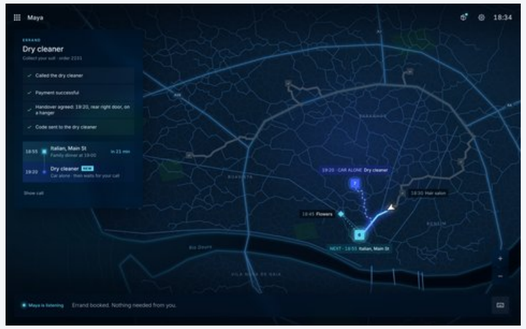
Courier mode: when nobody is in the car, it runs errands on its own.

In the scenario, a message says Michael's suit is ready at the dry cleaner. Maya offers to collect it once the family is at the restaurant. Afterwards the car charges nearby and waits until Michael calls it, with the suit inside.



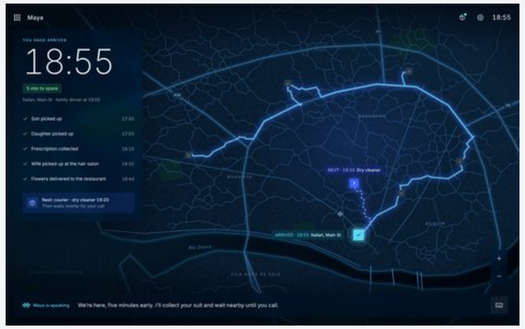
S18 · Errand suggestion

A message says the suit is ready. Maya offers to collect it with the empty car.



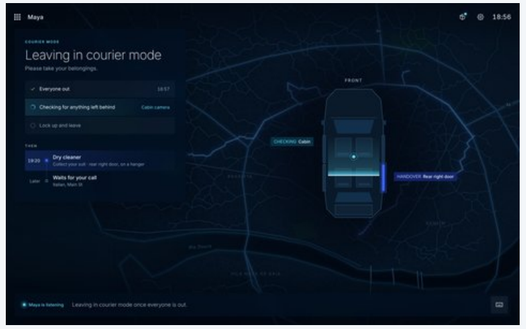
S19 · Errand booked

She calls, pays and agrees the handover. Only then is the errand booked.



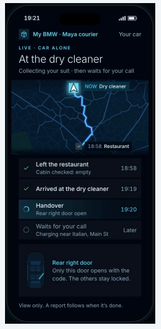
S20 · Arrival report

The arrival report ends with what happens next: the errand, then the car waits for his call.



C1 · Leaving in courier mode

The car checks that the cabin is empty and nothing is left behind.



C2 · Phone: courier run

Michael follows the run on his phone, view only.



C3 · Phone: errand report

A report with a photo, the receipt and the times.

THE RULES

Empty cabin only
Before leaving, the car checks for people, pets and forgotten items.

No handoff, no commitment
An errand is booked only once the shop confirms the time, the person and the handover.

One door, one code
Only the agreed door opens, once, with a one-time code; the cabin camera checks what goes in.

Ready when called
After the errand the car charges nearby; when Michael calls it, the phone says how soon it will arrive.

Every run is reported
Live on the phone, then a photo, the receipt and the times.

Charging is not an errand
It never appears in the plan. The car charges out of sight.

RISKIEST ASSUMPTIONS

A13 shops will hand items to an empty car · A14 people will send it, and accept a cabin camera · A15 an empty Level 5 car may legally run errands.

Nine rules every screen follows, in a car nobody has to drive.

1 Ask only what she can't know

As many questions as needed, no more. Schools, pickups and the prescription come from the profile and messages.

S06

2 The plan follows the conversation

Ambient intent: what Maya hears in the cabin or a call can reshape the plan.

S17

3 Speak only when needed

The screen is the confirmation. Maya never reads the plan back.

S07 · S12

4 Do first, show clearly, undo easily

No approval gates within the limits he sets, and no countdowns. Every action lands in "Done for you".

S09 · S14 · S17

5 The plan is alive

The world, the conversation or Michael can reshape it, and the ripple is visible.

S09 · S15 · S17

6 Spare time is the promise

Always visible. Green above 4 min, amber 0 to 4, red below 0, never colour alone.

S12 to S20 · V2

7 Central display to act, Panoramic Vision to glance

Panoramic Vision can't be touched and carries nothing unique.

Panoramic Vision

8 Voice and touch are equals

Every task works with either one alone.

S05 · S06 · S15

9 With you or without you

An empty car works for you, but only with an agreed handover.

S18 to C3

SAE DRIVING AUTOMATION LEVELS

0	1	2	3	4	5
No automation	Driver assistance	Partial	Conditional, takes over on request	High, in a defined area	Full, anywhere, no driver

WHY LEVEL 5

Only when nobody has to drive can an assistant use the whole trip, and can Michael talk, read and decide. One calm, trip-level experience instead of a design built around driver attention.

AT LEVEL 3 OR 4

The same concept could ship earlier, but every task would give way to a takeover request: Maya pauses and goes silent, and the central display and Panoramic Vision count down together (V1, p. 27).

HOW MICHAEL STAYS AWARE

- **The map is always there.** Position, the route ahead and what's done sit behind every screen.
- **The car states its intent first.** "Stop at School in 1 min", where it pulls in and which doors unlock (S13).
- **Every change shows what and why.** Removed stops crossed out, new ones marked, the old route dotted (S09, S17).
- **Car data has a fixed place.** Speed, temperature and range always hold the left half of Panoramic Vision.
- **Nothing happens out of sight.** Calls are transcribed, payments listed, recent actions can be undone.

Setting off: from sign-in to a plan Michael can trust at a glance.

Emotion: Michael's expected feeling, 1 low to 5 high · A2 to A16: assumptions, p. 28

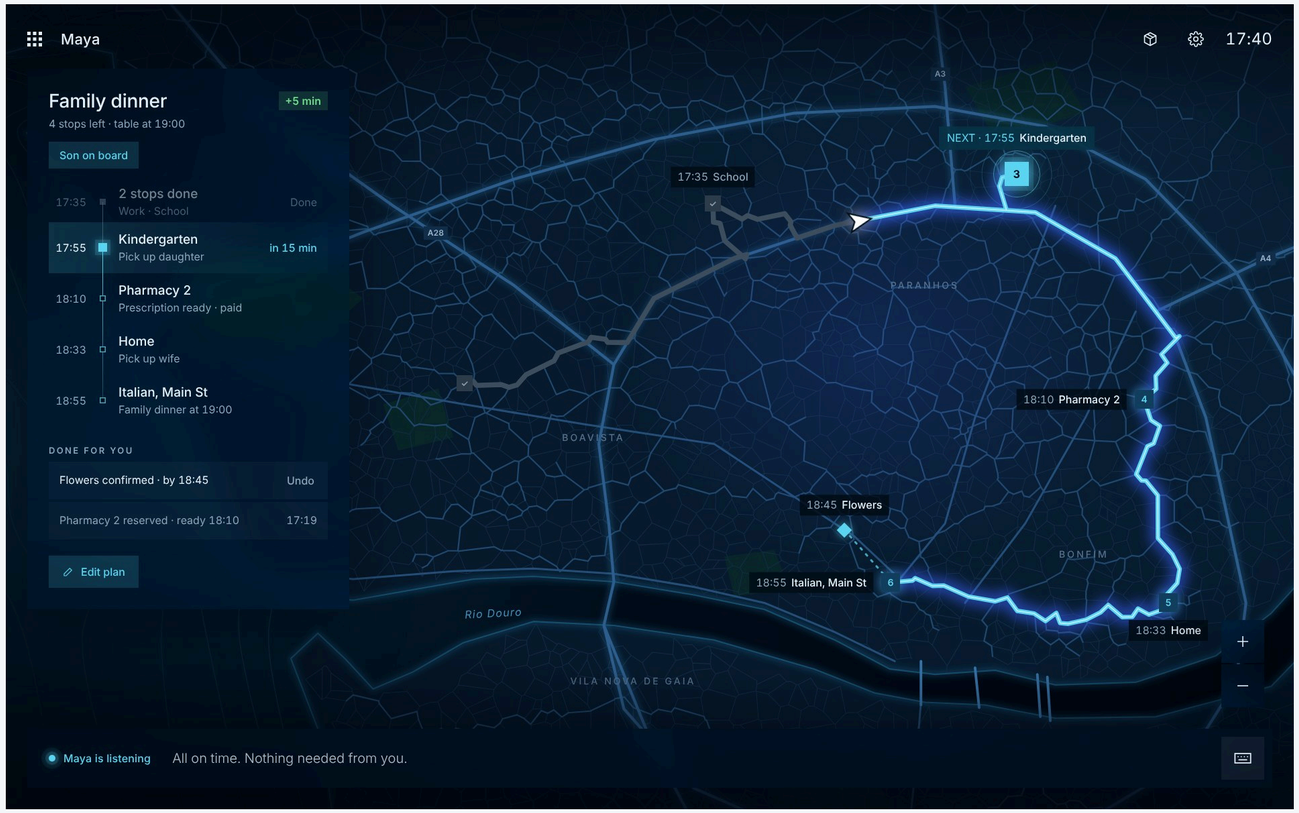
PHASE	Setting off				
TIME	17:15	17:16	17:19	17:22	17:24
BEAT	**Boarding and sign-in**	**Briefing**	**Pharmacy search**	**Flowers**	**The plan appears**
MICHAEL'S GOAL	Get going after work with no setup	Say what the evening needs, not where to go	Get the prescription as soon as possible	Not arrive empty-handed	Trust the plan at a glance
MICHAEL DOES OR SAYS	Gets in; signed in by face	"Family dinner tonight. Kids, my wife, and that prescription." Answers: the usual Italian on Main Street, at 19:00	Rides; may read the transcripts	"Yes. White roses."	Glances at it
MAYA SAYS	"Hi, I'm Maya, your concierge for the trip. How can I assist you?"	Asks only where, and what time	Silent	Suggests flowers once the pharmacy is secured, then silent	Silent
CENTRAL DISPLAY SHOWS	Sign in, then Welcome and "Going home" (S01, S02)	The "New plan" card fills in (S04 to S07)	Two calls with holds, then "Plan updated" (S08, S09)	Suggestion, then call, payment and "Delivery committed" (S10, S11)	Plan overview, spare time +5 min (S12)
PANORAMIC VISION SHOWS	◉ Home 17:48	◉ listening	Pharmacy changed · +5 min	◉ Calling Main St Florist, then "Flowers confirmed"	School 17:35 · +5 min
MAYA BACKSTAGE	Recognises Michael; predicts home from habit; loads calendar, contacts, messages	Parses needs; reads the prescription from messages; asks the gaps	Calls pharmacies on the route; pays; reroutes to pharmacy 2	Finds the birthday; picks an open florist near the restaurant; calls and pays from Maya Wallet	Builds the route; checks the range; arrival 18:55; replaces the default route
EXTERNAL	none	none	Pharmacy 1: no stock. Pharmacy 2: ready in 10 min	Florist agrees delivery by 18:45	none
EMOTION					
RISK	Wrong destination guess (A2); face not recognised	Misheard or missing detail (A6)	No pharmacy has stock (A4)	Florist refuses an AI caller (A4, A7)	Too much detail (A8)
DESIGN RESPONSE	The guess is a quiet default; PIN or password as fallback	The card is the confirmation; tap Maya's bar to type	Silent reroute with a visible banner	Self-introduction and live transcript	Spare time first, details on tap

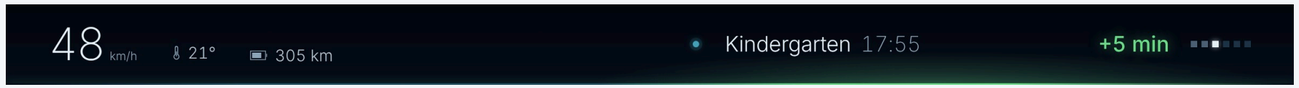
On the way, arrival, and the car alone.

The last three rows are the service blueprint: backstage, the outside world, and how the design answers each risk.

PHASE	On the way			Arrival	Car alone
TIME	17:33	18:03	18:32	18:55	18:58
BEAT	**Pickups**	**Wife’s call**	**The errand**	**Arrival**	**Courier run**
MICHAEL’S GOAL	Collect the kids without managing it	Absorb a change while on the phone	Get the suit without another trip	Arrive on time	Know the errand is done without watching it
MICHAEL DOES OR SAYS	Rides	Takes the call: “Sure, we’ll be there.”	“Yes, please.”	Steps out with the family	At dinner; glances at the phone
MAYA SAYS	Silent	Silent, chime only	Offers to collect the suit with the empty car after the drop-off	“We’re here, five minutes early. I’ll collect your suit and wait nearby until you call.”	Silent; notifications only
CENTRAL DISPLAY SHOWS	What the car is about to do, pickup states, “Done for you” (S13, S14)	Heard phrase, crossed and new stops, Undo (S16, S17)	Errand suggestion, then call, payment, agreed handover and a new plan row (S18, S19)	Arrival report, then “Leaving in courier mode” (S20, C1)	Central display off; the phone shows the live run and the report (C2, C3)
PANORAMIC VISION SHOWS	Next stop updates	Pickup moved: hair salon · +5 min	◉ Calling the dry cleaner, then “Errand booked: dry cleaner”	Arrived 18:55 · +5 min, then “Courier · waits for you”	Off (cabin empty)
MAYA BACKSTAGE	Tracks pickups and ETAs; tells the car where to stop and which doors to unlock	Hears the call; infers the change; replans	Reads the dry cleaner’s message; checks the car is free after 18:55; calls, pays and agrees the handover; texts a one-time code	Confirms arrival; checks the cabin is empty and nothing is left behind	Drives to the dry cleaner; opens one door for the code; the camera checks the suit; reports; charges near the restaurant until Michael calls
EXTERNAL	School and kindergarten release the children	Wife asks to be picked up at the hair salon	Dry cleaner agrees to hang the suit in the rear right door at 19:20	Flowers delivered 18:44; prescription collected 18:10	Staff type the code and hang the suit at 19:20
EMOTION					
RISK	Child not at the pickup point	Ambient false positive (A3, A9)	Shops won’t serve an empty car (A13); trust in sending it (A14)	Someone or something left in the cabin (A14)	Shop not ready or wrong item (A13, A14)
DESIGN RESPONSE	The intent card says how long the car waits	Chime, heard phrase, Undo with no countdown	No handoff, no commitment: booked only once the handover is agreed	Courier mode starts only with an empty cabin	One door, one code; a report with photo, receipt and times

A central display to act, Panoramic Vision to glance, and the phone only when the car is alone.





CENTRAL DISPLAY · ABOUT 15″, 1920 × 1200

Everything interactive. 16:10, seen from about 70 cm. The plan on the left, the live map behind, Maya's bar at the bottom.

PANORAMIC VISION · BASE OF THE WINDSCREEN

Passive and always dark. Car data on the left, Maya on the right. A copy of the central display, never the only channel. Off when the cabin is empty.

CHANNELS

V — Voice
Always listening in the cabin, no wake word. Only Michael's voice gives commands; other voices are context.

T — Touch
Everything on the central display: tap, drag, type. Every tap area is at least 76 × 76 px.

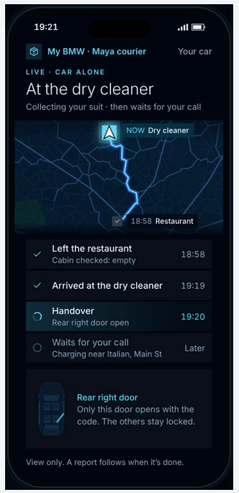
T — Keyboard
Tap Maya's bar to type. A large keyboard takes the lower half of the screen.

S — Captions
Everything Maya says appears in her bar, so nothing is audio only.

S — Chime
One soft sound, only for changes heard in conversation, always with the heard phrase on screen.

S — Light
The horizon line of Panoramic Vision carries the spare-time colour and flashes for 4 s on events.

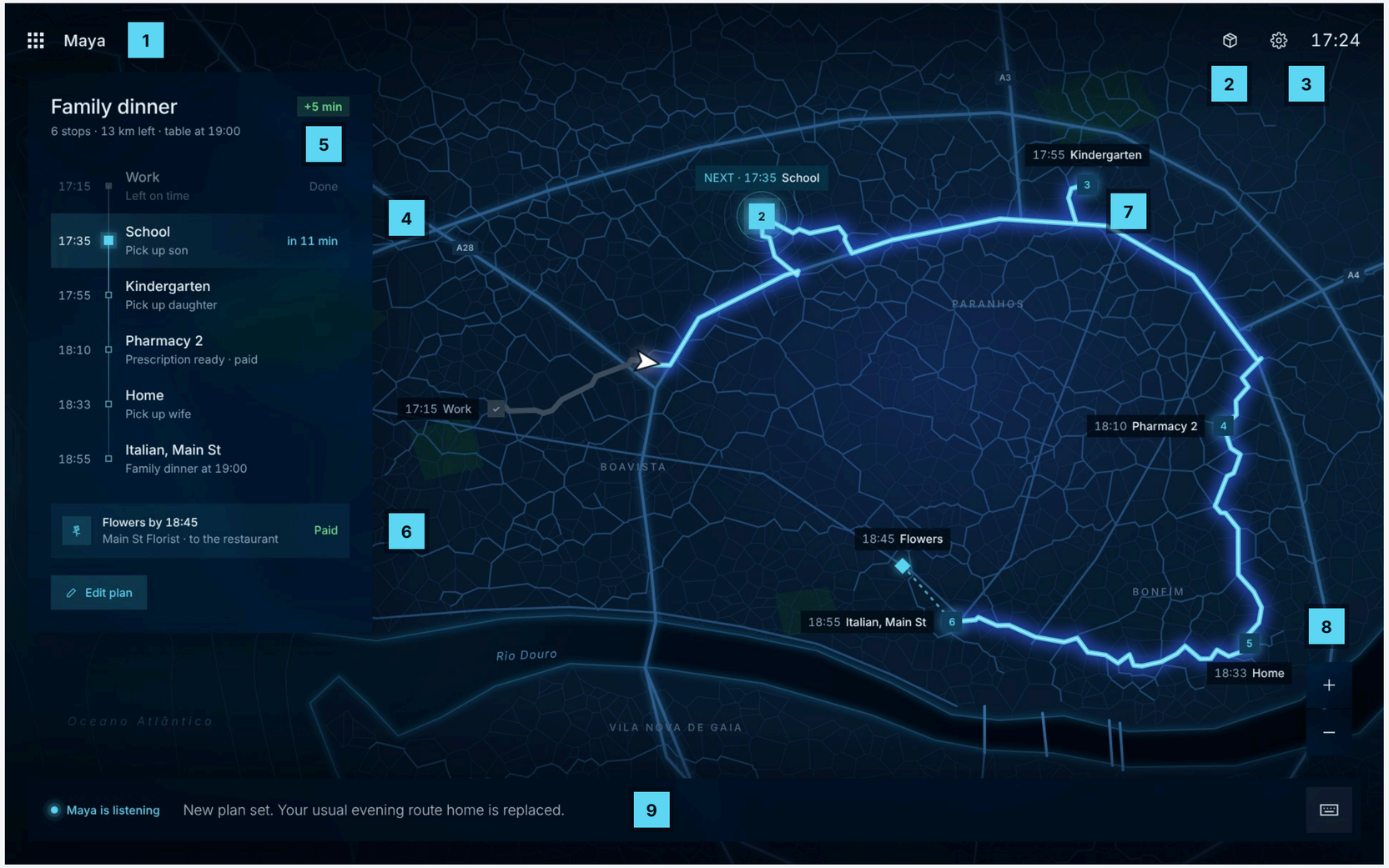
V voice T touch S system output



PHONE

Courier mode only, and view only, in the My BMW app: the live run (C2) and the report (C3).

One layout everywhere: the plan on the left, the map behind, Maya below.



1 App launcher and Maya
Other apps sit behind the launcher. “Maya” names what is on screen.

2 Errands
Opens Errands (C4). A dot shows while an errand is booked or running.

3 Settings and clock
Nothing else in the top bar.

4 Plan panel
Trip title, stops left, distance and the table time; the stop timeline, next stop highlighted; Edit plan. Glass over the map.

5 Spare time
The promise, always visible: +5 min in green.

6 Done for you
What Maya arranged: flowers by 18:45, paid.

7 Live map
The whole trip. The route ahead glows, the travelled route is grey; square pins; sonar rings on the next stop.

8 Zoom
In and out, aligned with the keyboard button.

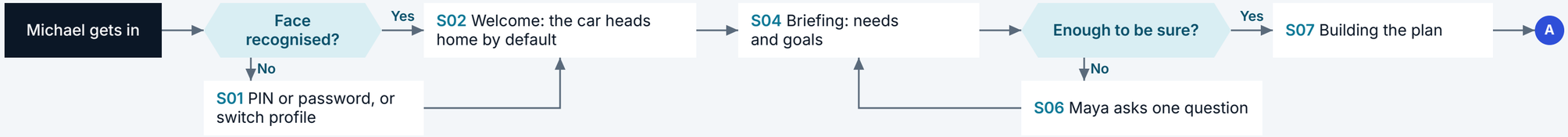
9 Maya’s bar
Her status (listening, thinking, speaking, on a call), her words as a caption, and the keyboard.

The map is Porto, drawn from its real coast, river and main roads; the minor streets are illustrative.

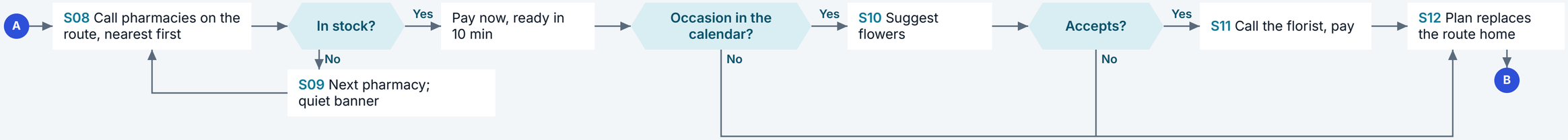
Every decision behind the evening, in four lanes.

step · decision · continues · back to the trip

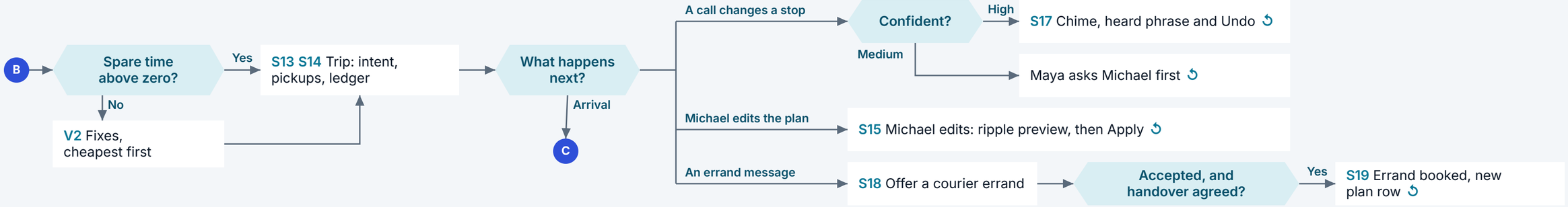
1 · GETTING GOING



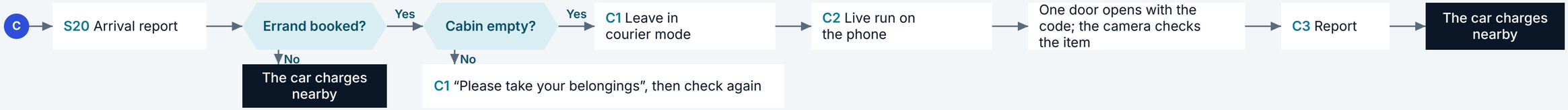
2 · PLANNING



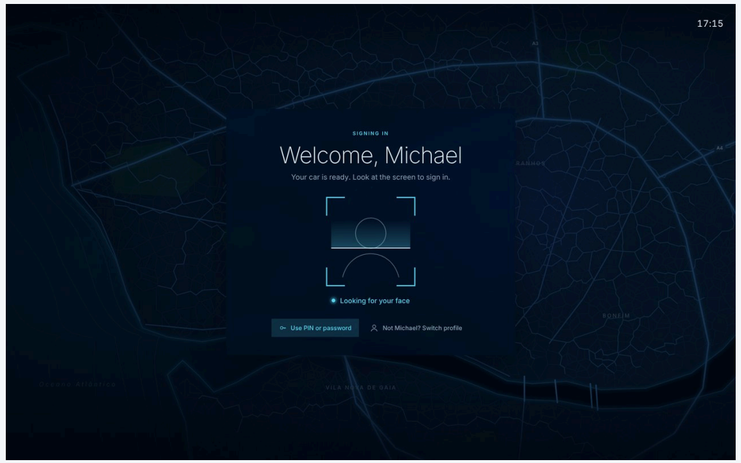
3 · ON THE WAY



4 · AFTER ARRIVAL

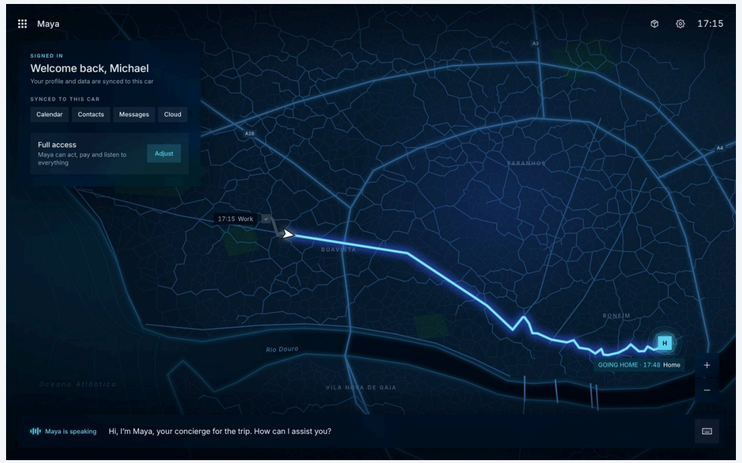


The car expects Michael, signs him in by face and sets off towards home.



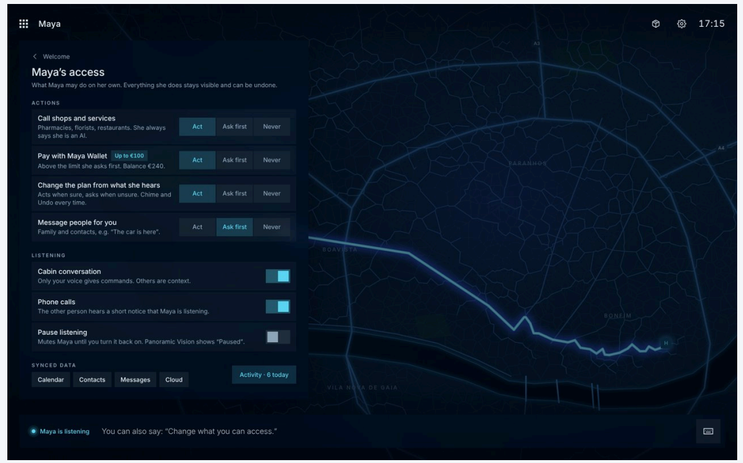
S01 Sign in

- The car expects Michael but signs him in only once it recognises his face.
- Fallbacks: Use PIN or password, or “Not Michael? Switch profile”.
- Before sign-in the top bar shows only the clock.



S02 Welcome

- “Welcome back, Michael”: calendar, contacts, messages and cloud synced to this car.
- Already moving on the most probable route: Going home, 17:48.
- “Full access” sums up Maya’s rights, with Adjust.



S03 Maya’s access

- Opened from Adjust. Per action: Act, Ask first or Never.
- Maya Wallet pays on its own up to €100; above that, Maya asks first.
- Listening: cabin, phone calls (the other person hears a notice), pause.

SPOKEN

MAYA	“Hi, I’m Maya, your concierge for the trip. How can I assist you?”
TOUCH TWIN	Adjust opens Maya’s access; by voice, “Change what you can access.”

PANORAMIC VISION · MAYA’S HALF



Home 17:48. No plan yet, so spare time is not shown and the line stays cyan.

Michael says what the evening needs. Maya asks only what she can’t know.

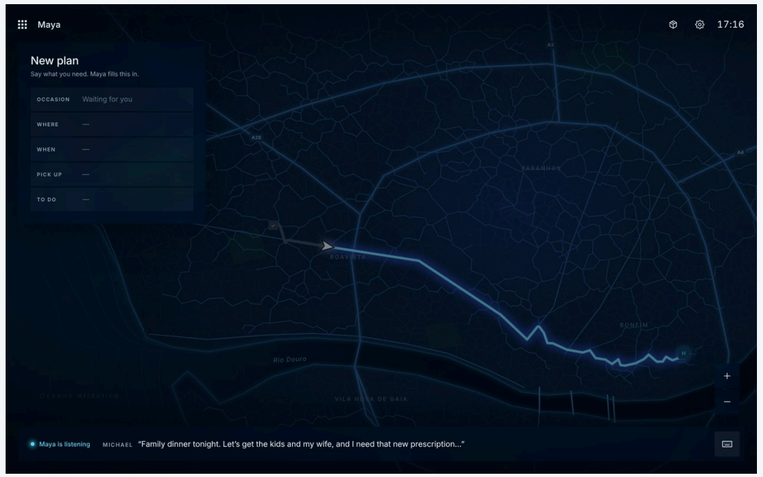
SPOKEN

MICHAEL	“Hi Maya. Family dinner tonight. Let’s get the kids and my wife, and I need that new prescription from my messages as soon as possible.”
MAYA	“Where is the dinner?”
MICHAEL	“The usual Italian on Main Street.”
MAYA	“What time?”
MICHAEL	“Seven.”
TOUCH TWIN	Type in Maya’s bar (S05) and answer in the search field (S06).

PANORAMIC VISION · MAYA’S HALF

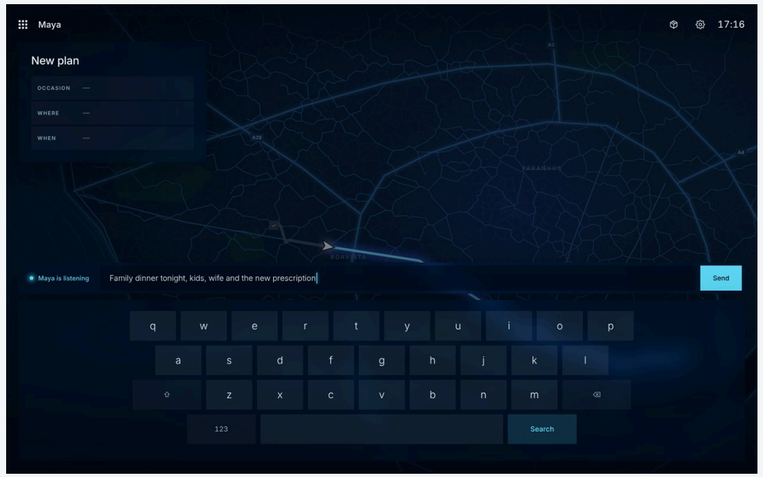


The presence dot brightens while Maya hears Michael. Home 17:48.



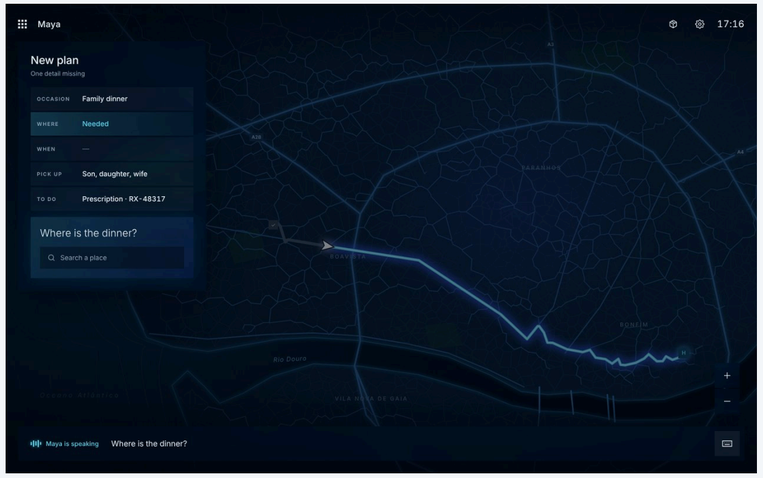
S04 Briefing: listening

The “New plan” card: occasion, where, when, pick up, to do. It fills in as Michael talks.



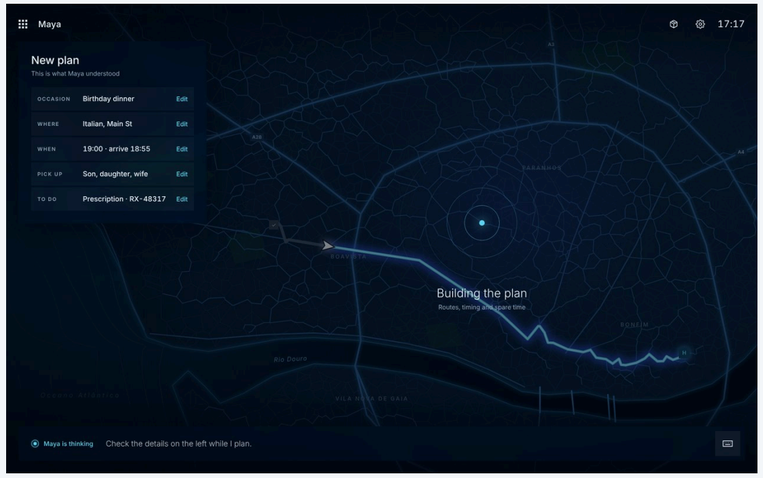
S05 Briefing: keyboard

Touch twin: tap Maya’s bar to type. The keyboard takes the lower half; the card shrinks.



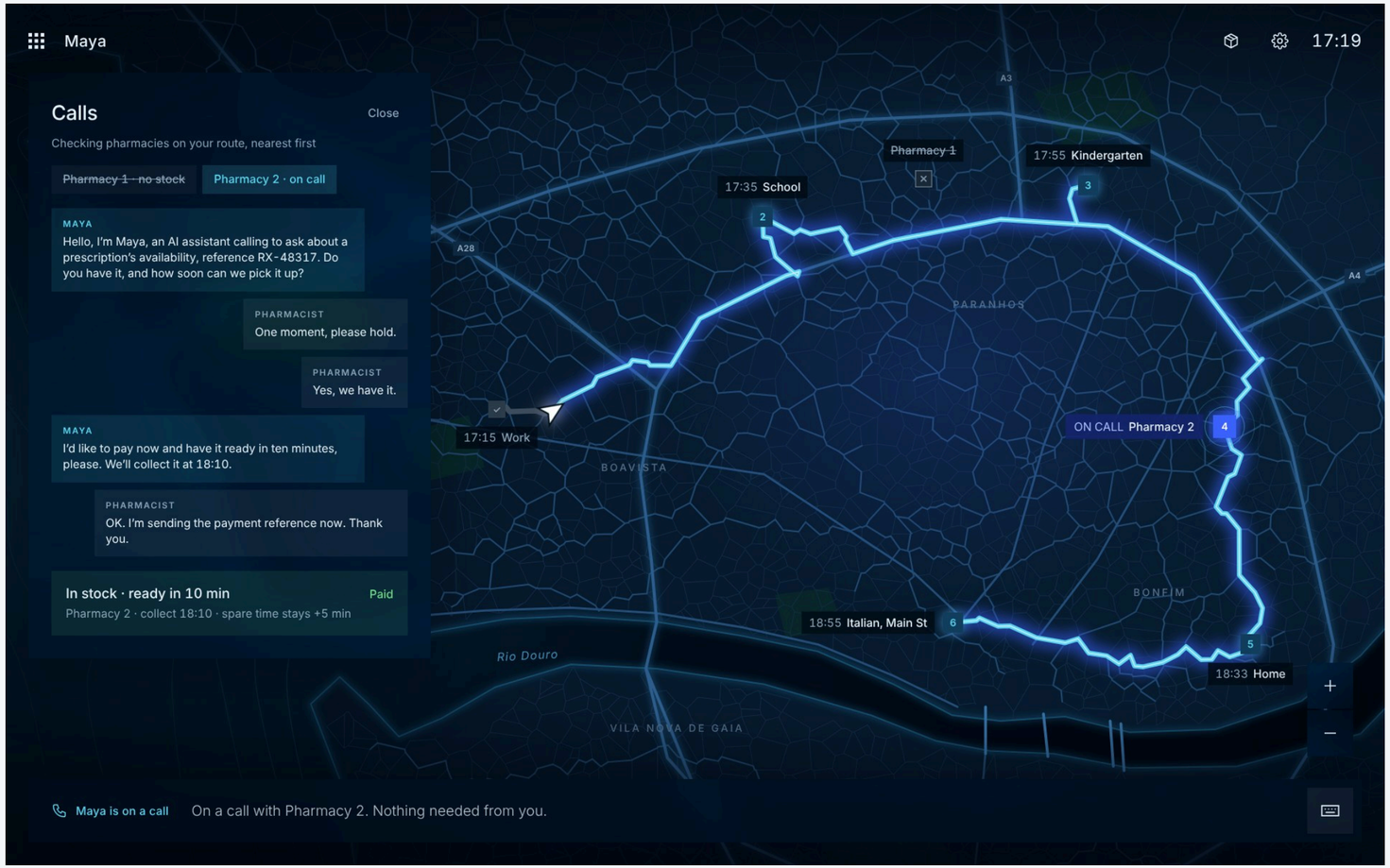
S06 Briefing: Maya asks

Maya knows the pickups and the prescription. She asks only where, then when. No suggested answers.

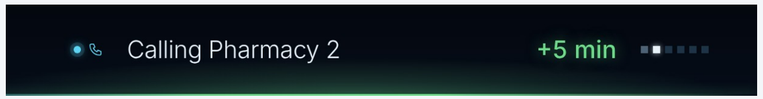


S07 Briefing: building the plan

“This is what Maya understood”, all still editable, while she builds the plan. No read-back.



PANORAMIC VISION · MAYA'S HALF



On a call: dot and phone. The plan now exists, so spare time appears: +5 min.

SPOKEN

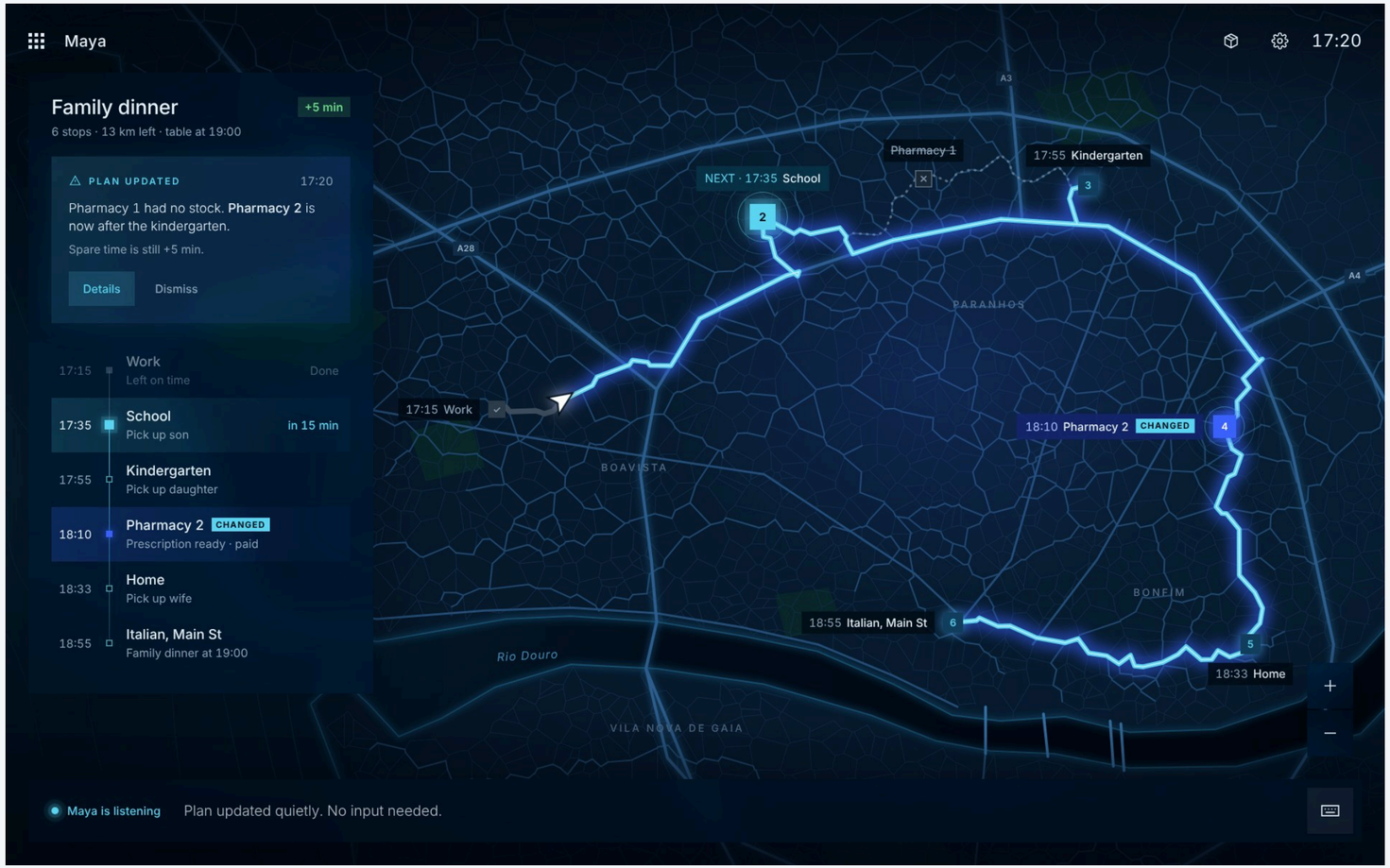
MAYA	"Hello, I'm Maya, an AI assistant calling to ask about a prescription's availability, reference RX-48317."
PHARMACIST	"Yes, we have it."
MAYA	"I'd like to pay now and have it ready in ten minutes, please."

Maya phones pharmacies on the route, nearest first. The cabin stays quiet.

S08 Retailer calls

- Calls are silent in the cabin and transcribed live. Maya always says she is an AI.
- Pharmacy 1 has no stock and is struck through. Pharmacy 2 puts her on hold, then confirms.
- Paid from Maya Wallet, ready in 10 min, collected at 18:10. Spare time stays +5 min.
- On the map, Pharmacy 1 is crossed out and Pharmacy 2 marked "on call".
- The shared screen shows the prescription's reference number, never the medication.
- Maya uses a shop's own ordering system when it has one; phone calls are the fallback, because most local shops have none.

TOUCH TWIN Close hides the calls panel. Nothing is needed from Michael.



PANORAMIC VISION · MAYA'S HALF



A 4-second event: "Pharmacy changed +5 min". No chime; the line flashes white.

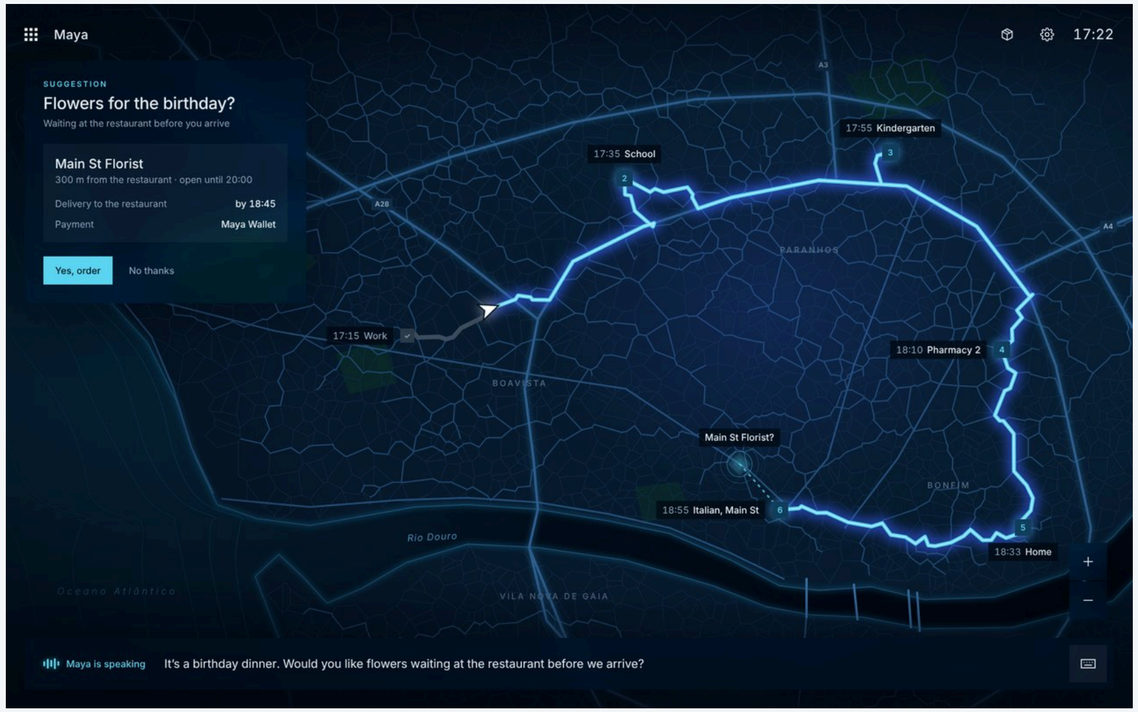
The world changed, so the plan changes quietly, with no input from Michael.

S09 Plan change: automatic

- A change in the world, made silently: Pharmacy 1 had no stock, so Pharmacy 2 now comes after the kindergarten.
- No chime, no input. The banner says what changed and why, and stays until dismissed.
- Pharmacy 2 is marked CHANGED in the list and on the map; the old route stays dotted.
- Spare time is still +5 min.

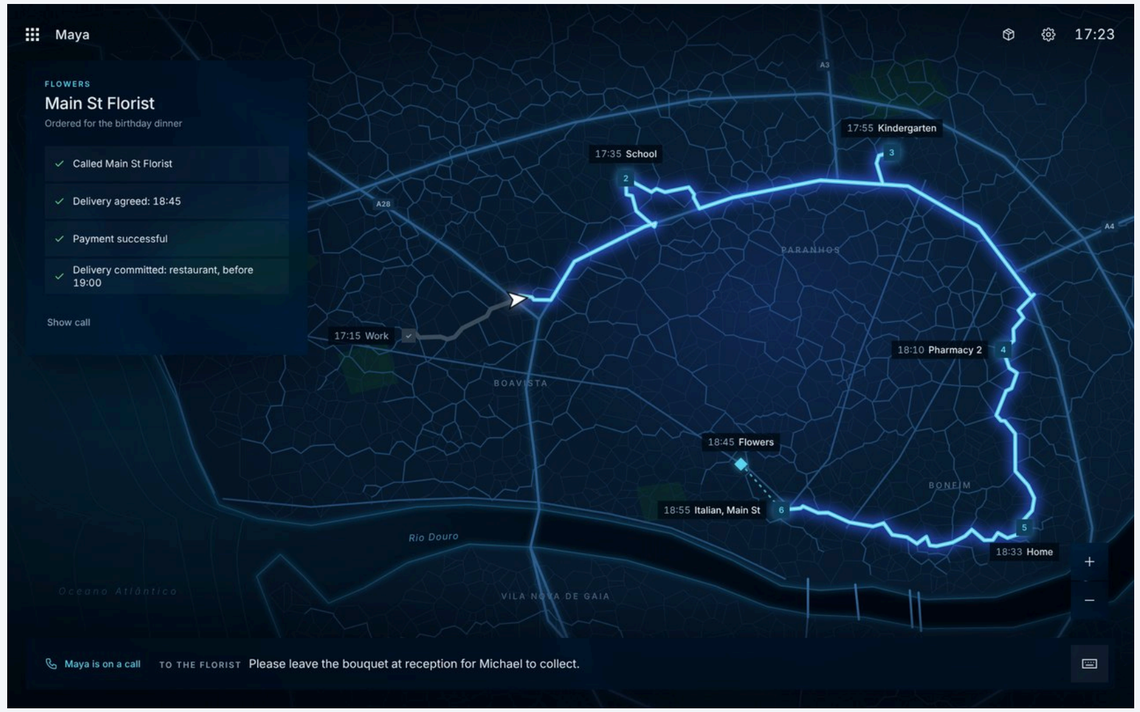
TOUCH TWIN Details reopens the calls; Dismiss closes the banner.

A birthday in the calendar becomes a suggestion, then a call becomes a gift.



S10 Gift suggestion

- Offered once the pharmacy is secured. One florist, no bouquet choices until Michael says yes.
- Main St Florist, 300 m from the restaurant: delivery by 18:45, paid with Maya Wallet.
- Until he answers, the florist is a dotted pin on the map.



S11 Florist order

- Four confirmed steps: called, delivery agreed for 18:45, payment successful, delivery committed before 19:00.
- Show call opens the transcript; Maya's last words to the florist stay in her bar.
- The pin turns solid, with a dotted delivery line to the restaurant.

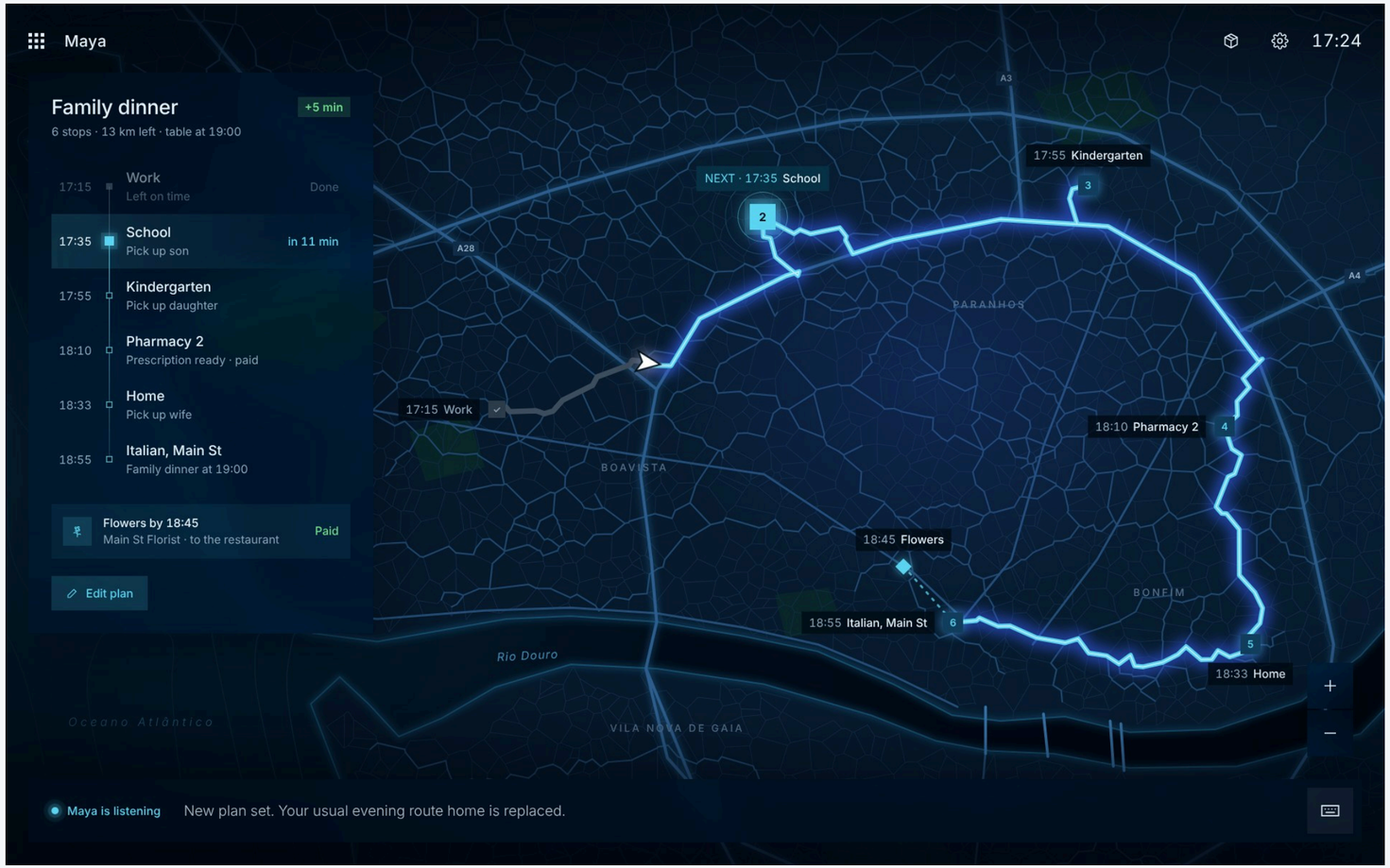
SPOKEN

MAYA	"It's a birthday dinner. Would you like flowers waiting at the restaurant before we arrive?"
MICHAEL	"Yes. White roses."
TOUCH TWIN	Yes, order or No thanks.

PANORAMIC VISION · MAYA'S HALF



Speaking: rings around the dot. School 17:35, +5 min.



PANORAMIC VISION · MAYA'S HALF



The resting state: next stop, spare time and one square per stop.

SPOKEN

MAYA'S BAR "New plan set. Your usual evening route home is replaced."

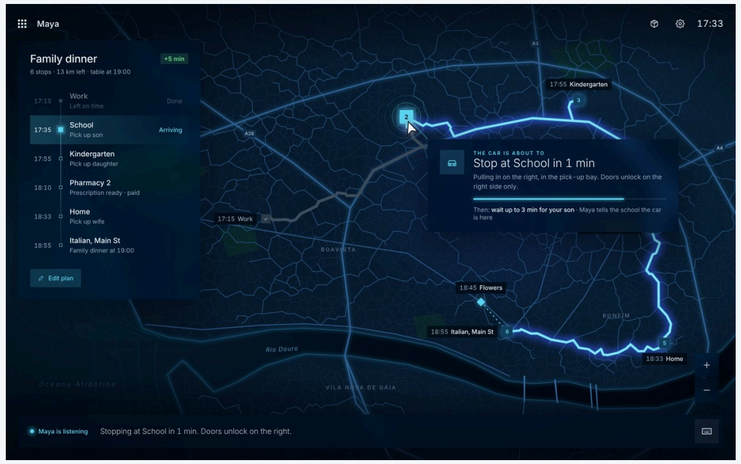
The route home is replaced by the plan. No speech: the screen is the confirmation.

S12 Plan overview

- Family dinner: 6 stops, 13 km left, table at 19:00, and +5 min of spare time at the top.
- The stop timeline: Work done and greyed, School next with "in 11 min".
- Flowers by 18:45, paid, in their own row; Edit plan below.
- The whole trip on the map: the route ahead glows, the travelled route is grey, the next stop has sonar rings.
- The car never stopped moving: the plan simply takes over from "Going home".
- Maya checks the range as she builds the plan: about 15 km of driving, well within it.

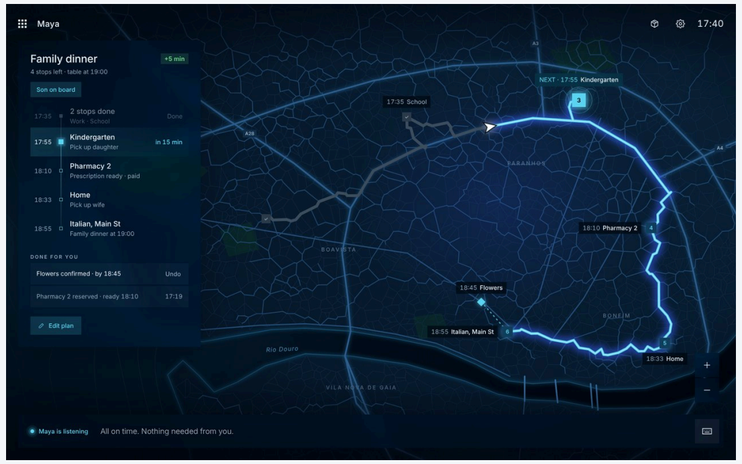
TOUCH TWIN Tap a stop, Edit plan (S15), zoom. By voice: "What's next?"

The car says what it is about to do. Maya keeps a ledger of what she did.



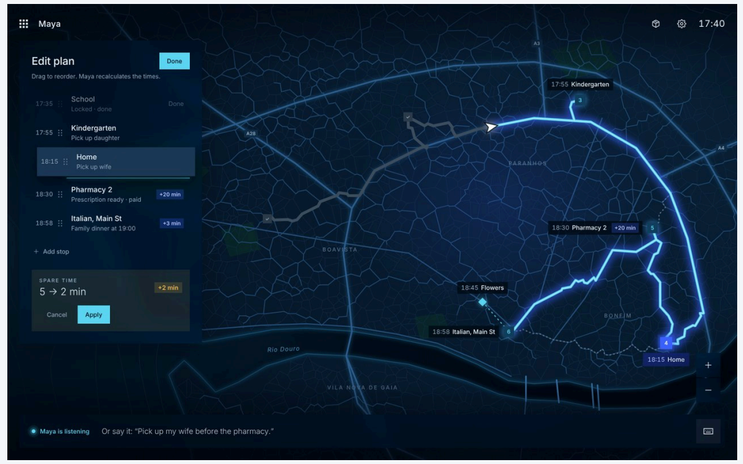
S13 Vehicle intent

- About a minute before each stop: where the car pulls in and which doors unlock.
- Then: wait up to 3 min; Maya tells the school the car is here.
- The stop row reads “Arriving”.



S14 Active trip

- “Son on board”; finished stops fold into one row.
- “Done for you”: the last two actions, with Undo on the latest only.
- Edit plan is always one tap away.



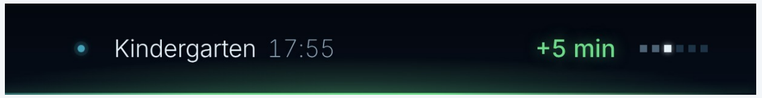
S15 Edit plan

- A branch: drag to reorder. The ripple shows every later time, +20 min, +3 min.
- Spare time 5 → 2 min, in amber. Nothing changes until Apply.
- The map previews the new route; the old one is dotted.

SPOKEN

MAYA’S BAR	“Stopping at School in 1 min. Doors unlock on the right.”
MAYA’S BAR	“All on time. Nothing needed from you.”
TOUCH TWIN	Edit plan, or say “Pick up my wife before the pharmacy.”

PANORAMIC VISION · MAYA’S HALF



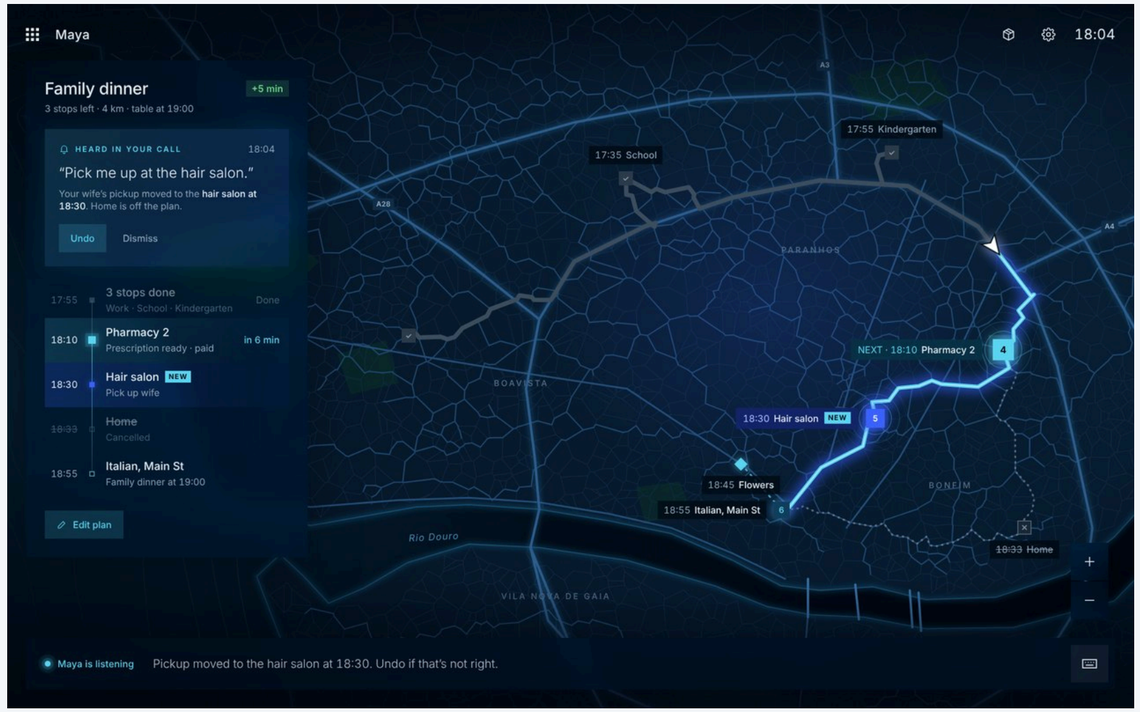
The next stop updates as stops complete: Kindergarten 17:55, +5 min.

Maya hears "pick me up at the hair salon" and moves the stop, with a chime and Undo.



S16 Incoming call

- A call card over the map: Wife, Mobile, Answer and Decline. The plan stays visible behind it.
- Maya listens quietly; the other person hears a short notice that she is listening.



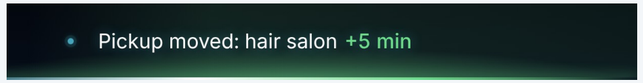
S17 Plan change: ambient

- "Heard in your call": the phrase, what changed, then Undo and Dismiss. No countdown.
- Hair salon marked NEW at 18:30; Home crossed out; the old route dotted.
- One soft chime, no speech. If Maya were unsure, she would ask instead.

SPOKEN

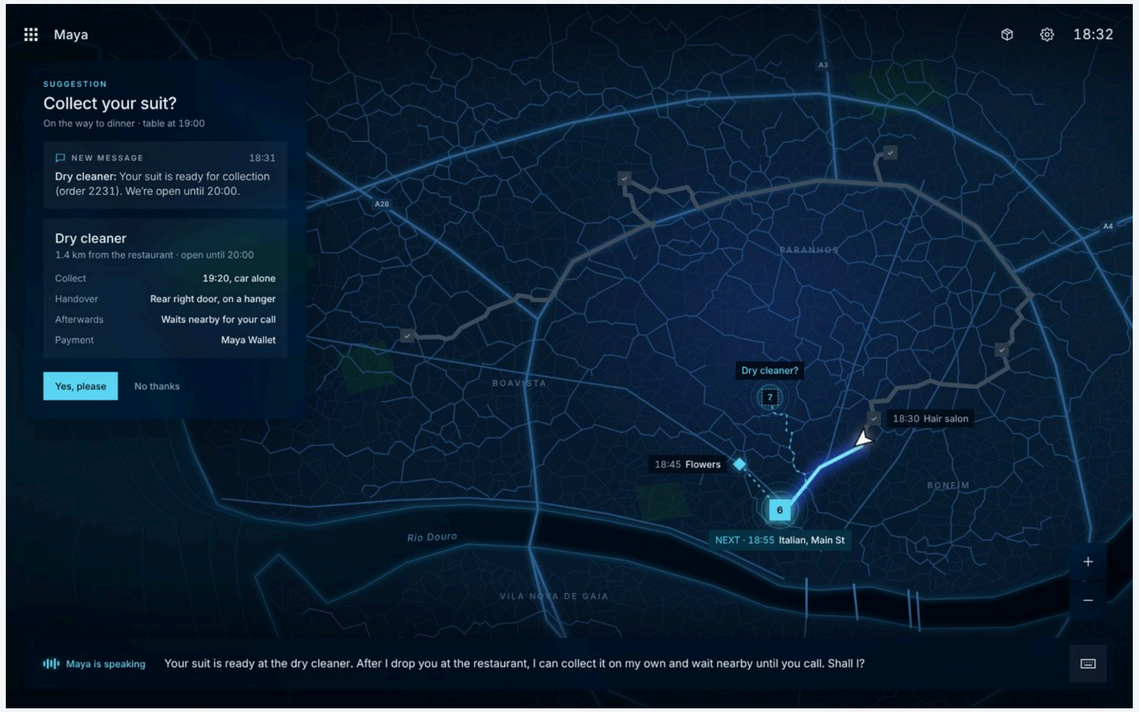
WIFE	"Change of plan: pick me up at the hair salon, not at home."
MICHAEL	"Sure, we'll be there."
TOUCH TWIN	Undo, or change the pickup in Edit plan (S15).

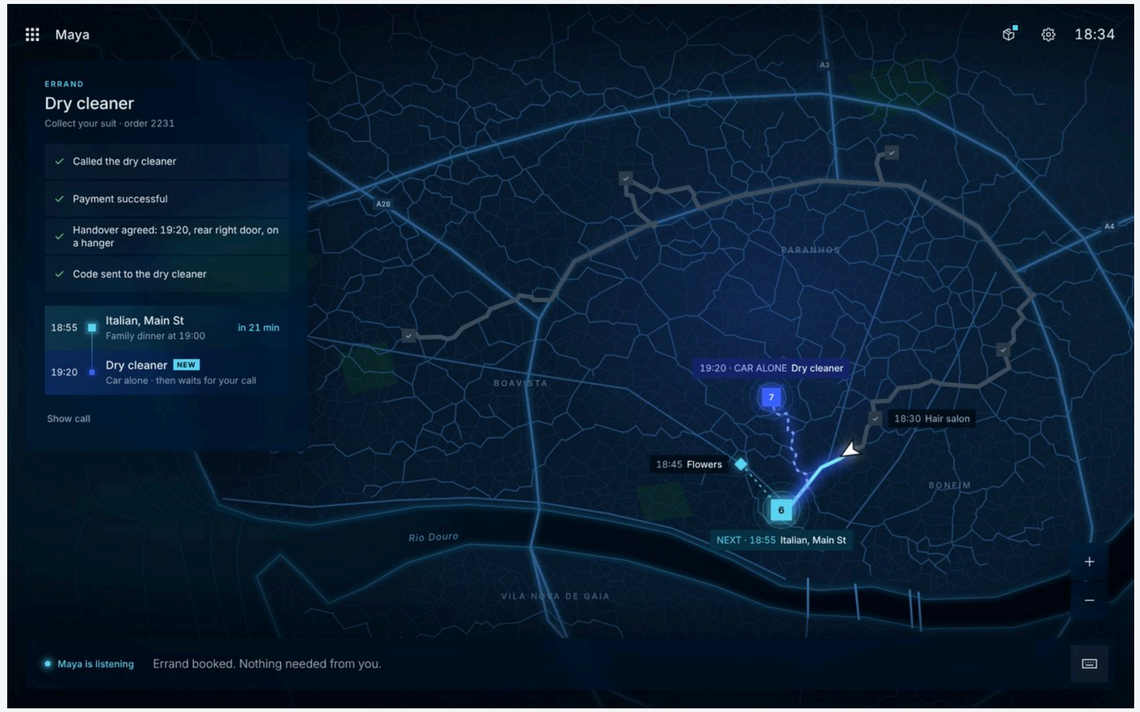
PANORAMIC VISION · MAYA'S HALF



A 4-second event: "Pickup moved: hair salon +5 min". The line flashes white.

A message from the dry cleaner becomes an errand for the empty car.





S18 Errand suggestion

- The dry cleaner's message is shown, not read aloud.
- The offer: 19:20, car alone, rear right door, then it waits nearby for his call; paid with Maya Wallet.
- The dry cleaner is a suggested pin until Michael answers.

S19 Errand booked

- Booked only once the handover is agreed: called, paid, handover agreed, code sent.
- A new plan row after the restaurant: 19:20 Dry cleaner, car alone, then waits for his call.
- A dot on the Errands icon while an errand is booked or running.

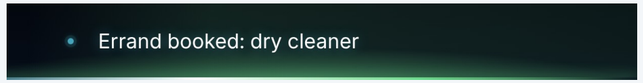
SPOKEN

MAYA "Your suit is ready at the dry cleaner. After I drop you at the restaurant, I can collect it on my own and wait nearby until you call. Shall I?"

MICHAEL "Yes, please."

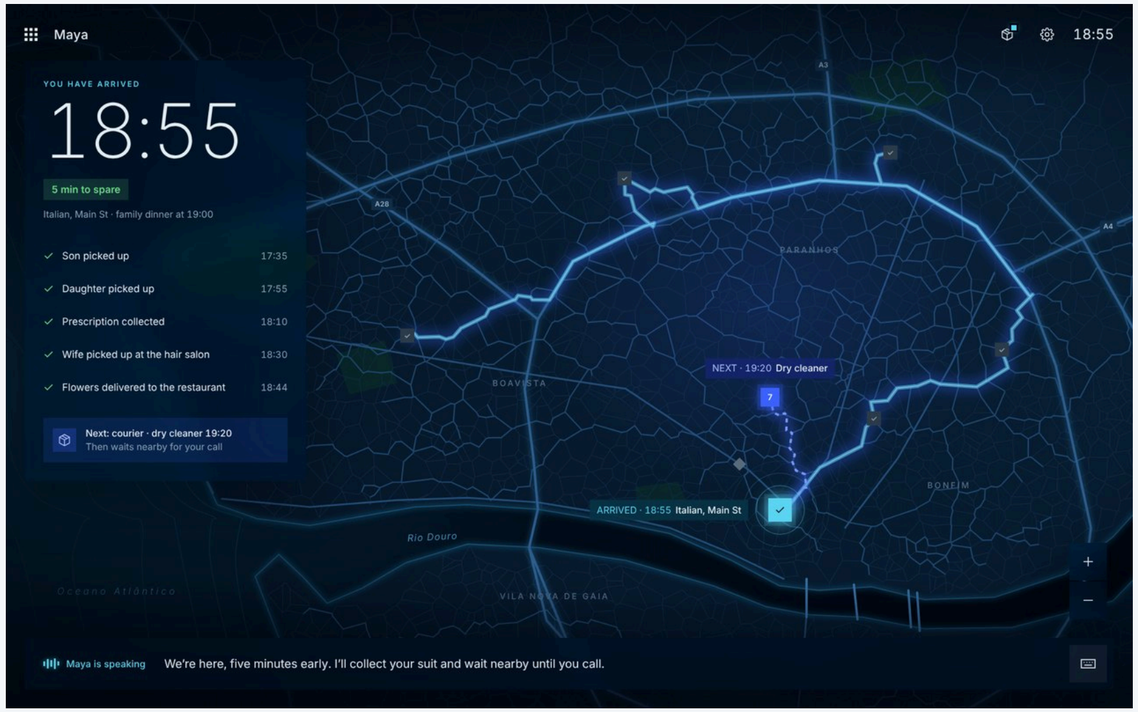
TOUCH TWIN Yes, please or No thanks.

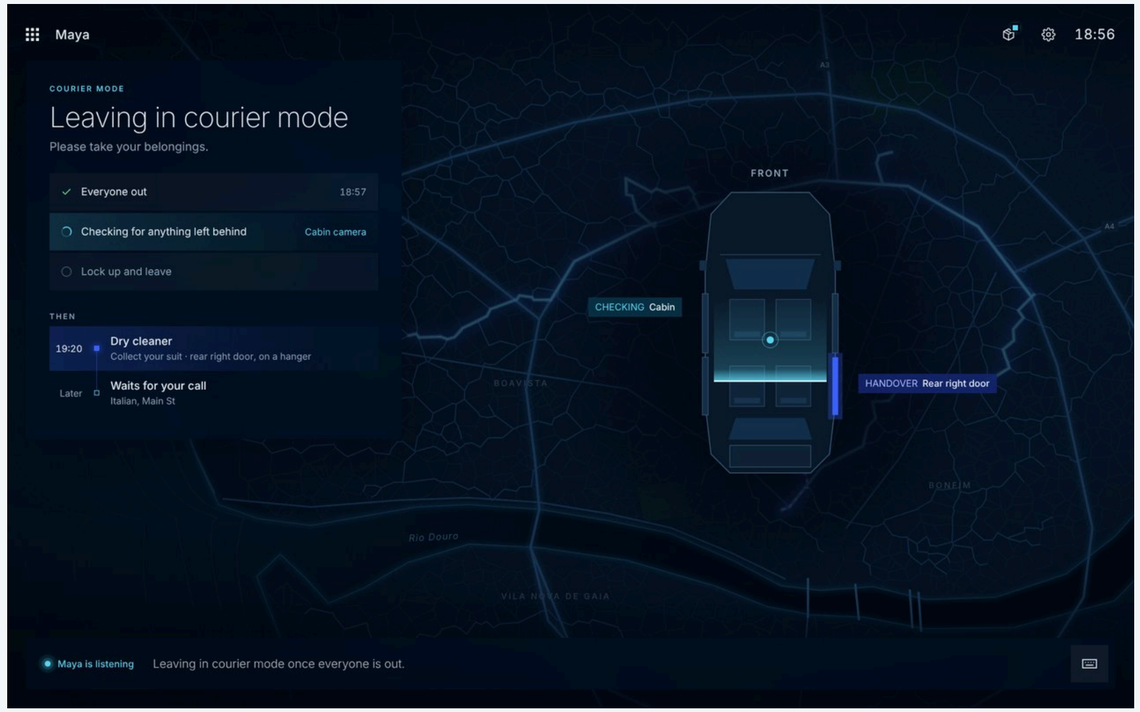
PANORAMIC VISION · MAYA'S HALF



"Errand booked: dry cleaner" for 4 seconds. Spare time is unchanged.

Five minutes early. Then the car checks it is empty and leaves on its errand.





S20 Arrival report

- 18:55 and 5 min to spare, readable in one glance.
- Everything done, with times: son, daughter, prescription, wife at the hair salon, flowers delivered 18:44.
- Next: courier, dry cleaner 19:20, then the car waits nearby for his call.

C1 Leaving in courier mode

- "Please take your belongings."
- Everyone out, then the cabin camera checks for anything left behind, then lock up and leave.
- A top view of the car shows the cabin check and the handover door.

SPOKEN

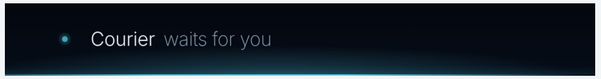
MAYA "We're here, five minutes early. I'll collect your suit and wait nearby until you call."

PANORAMIC VISION · MAYA'S HALF

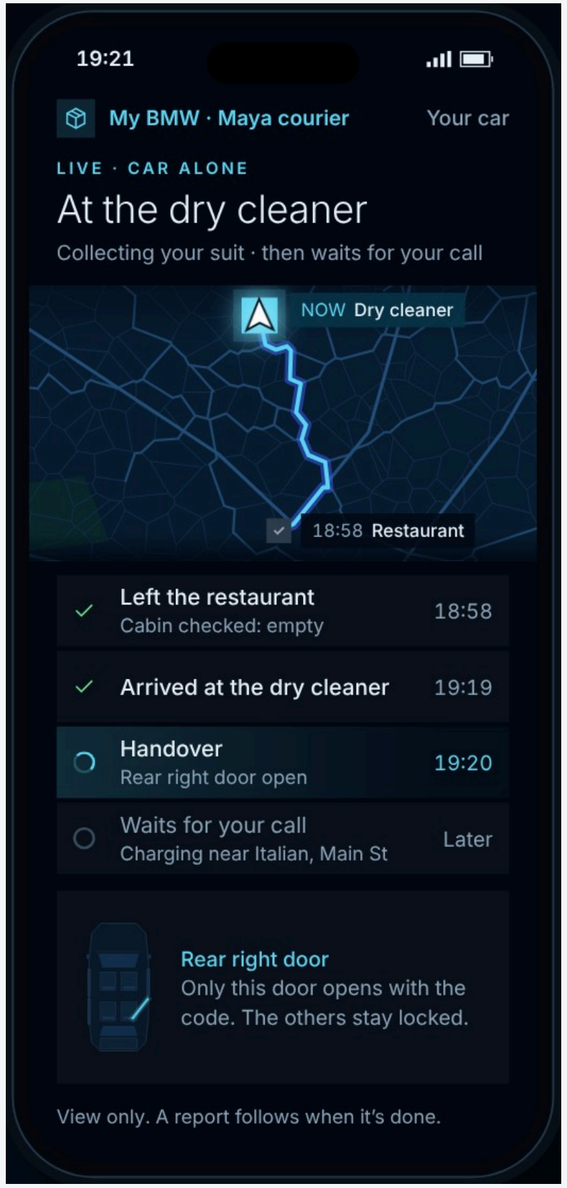


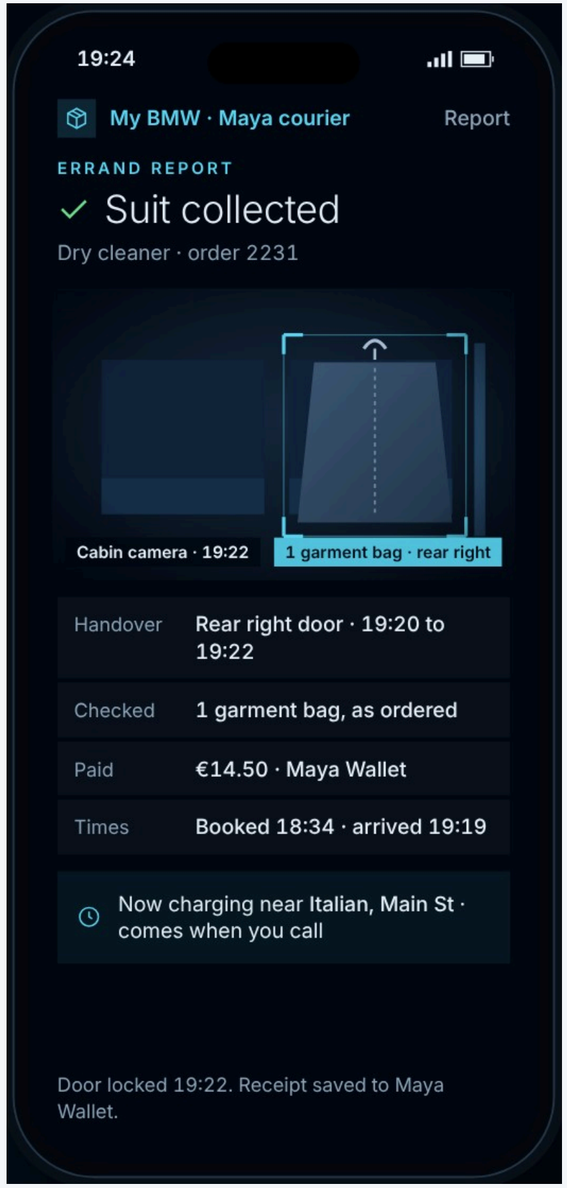
Arrived 18:55, every square filled, +5 min.

THEN



"Courier · waits for you" while the family gets out.





Michael follows the run on his phone, view only, and gets a report.

C2 Phone: courier run

- Live, with the car alone, in the My BMW app: the map and the steps. Left the restaurant 18:58, arrived 19:19, handover 19:20, then it waits for his call.
- Which door is in use: only the rear right door opens with the code; the others stay locked.
- View only: no controls and no voice.

C3 Phone: errand report

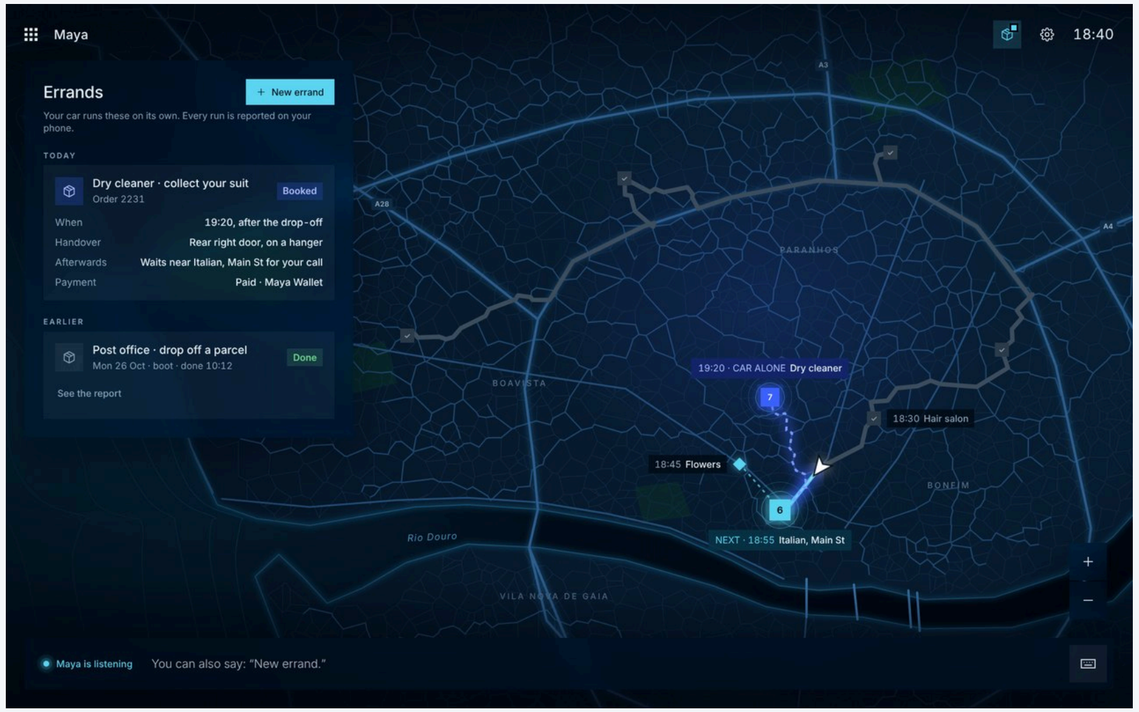
- A still from the cabin camera: 1 garment bag, rear right.
- Handover 19:20 to 19:22, checked as ordered, paid €14.50, booked 18:34, arrived 19:19.
- Now charging near the restaurant until he calls. Door locked 19:22, receipt saved to Maya Wallet.

WHERE THE STORY ENDS

The car drives to a charging station near the restaurant. Charging is not an errand, so it never appears in the plan. The same car, with the suit inside, waits there until Michael calls it.

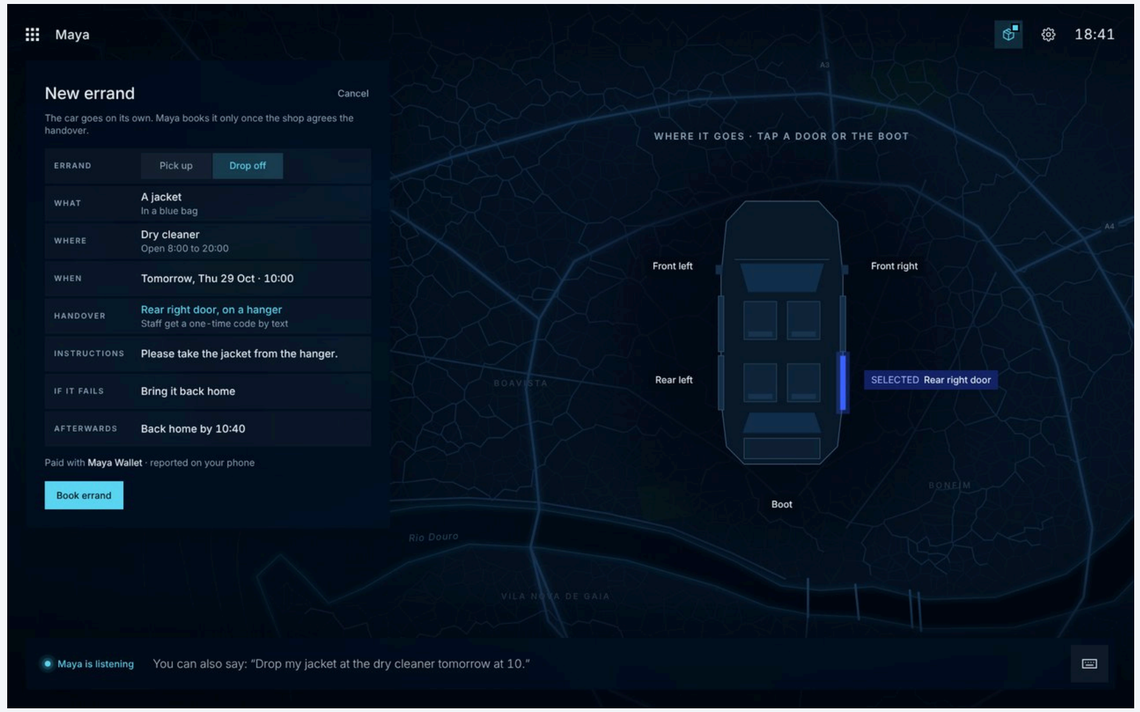
DISPLAYS Off: the cabin is empty, so the central display and Panoramic Vision are off.

Errands can also be booked by hand, from the box icon beside Settings.



C4 Errands

- Today: the dry cleaner, Booked, with when, handover, what happens afterwards and payment.
- Earlier: Post office, drop off a parcel, Mon 26 Oct, boot, Done, with its report.
- New errand at the top.



C5 New errand

- Pick up or Drop off; what, where, when.
- Handover: tap a door or the boot on a top view of the car. Staff get a one-time code by text.
- Instructions, what to do if it fails, what the car does afterwards; then Book errand.

SPOKEN

VOICE TWIN "Drop my jacket at the dry cleaner tomorrow at 10."

Tokens verified against automotive guidance.



Abyss
Background

Glass
Panels, 30–60% over the map

Lumen
Primary accent

Cobalt
Glow, new items, errands

Green
More than 4 min spare

Amber
0 to 4 min spare

Red
Late, takeover

Text
Primary

Muted
Secondary

TYPE · INTER

Display	128 / ExtraLight	Arrival time (S20)
Hero	44–60 / Light	Sign-in, call card, courier title
Large	24–30 / Light	Clock, heard phrase, intent
Title	28 / Regular	Panel titles
Body	19–20 / Regular	Stops, Maya's captions
Secondary	16–17 / Regular	Times, details, notes
Label	13 / SemiBold caps	Kickers and row labels
Badge	12 / SemiBold caps	NEW, CHANGED, CHEAPEST

Dark first: the night cabin is the hard case, and light can carry meaning. Panoramic Vision is always dark. Material 3 and Android for Cars are references for sizes and behaviour only.

RULES

- **Sharp corners, no borders.** Fill, glass and spacing group things, never lines.
- **Glass over a full-bleed map.** Panels at 30 to 60% with a 24 px blur, so the map is always there.
- **76 × 76 px tap areas.** Visual sizes stay (48 to 64 px), but every tap area meets the Android for Cars minimum of 76 dp, at about 1 px per dp.
- **16 px minimum text.** Except uppercase labels (13 to 14 px) and small badges (12 px).
- **About 15″, 1920 × 1200, at about 70 cm.** A 16:10 central display, like today's large infotainment screens.
- **Colour is never alone.** Spare time always has a number; late adds a minus sign and a warning mark.
- **One sound.** A soft chime, only for changes heard in conversation.
- **Motion with a job.** Panoramic Vision events last 4 seconds; Maya's presence pulses slowly; nothing loops while she speaks.

Component inventory

38 components, drawn with the same code as the screens · every tap area is at least 76 × 76 px

Top bar — CHROME
App launcher · Maya wordmark · Errands button with badge · Settings · clock. Clock only before sign-in
All central display screens

Icon button — CHROME
Default · active · with badge. 48 to 64 px visual, 76 px tap area
All central display screens

Maya's bar — CHROME
Listening · thinking · speaking · on a call. Caption in one or two lines, with speaker label. Keyboard button; text-entry mode
S02 to S20, C1, C4, C5, V1, V2

Zoom control — CHROME
Zoom in · zoom out
Map screens

Route line — MAP
Live · travelled · previous plan · delivery · courier · suggested errand · traffic
S02 to S20, C1, C4, V2

Stop pin — MAP
Upcoming · next with sonar rings · done · new · cancelled · suggested · arrived
S02 to S20, C4, V1, V2

Delivery pin — MAP
Default · suggested · done
S10 to S20

Car marker — MAP
Moving, with heading
S02 to S19, C4, V2

Map label — MAP
Upcoming · next · new · done · cancelled · suggested. Place names and road shields
S02 to S20, C4, V2

Panel — CONTENT
Narrow · wide · form. Glass at 30 to 60 percent, 24 px blur, no border
S02 to S20, C1, C4, C5, V2

Spare-time pill — CONTENT
Green · amber · red with minus and warning mark · large
S09, S12 to S17, S20, V2

Stop row — CONTENT
Upcoming · next · arriving · done · summary · new · changed · cancelled · dragging
S09, S12 to S15, S17, S19, C1, V2

Briefing slot — CONTENT
Empty · asked · filled · editable
S04 to S07

Suggestion card — CONTENT
Florist · errand, with key–value rows and a quoted message
S10, S18

Change banner — CONTENT
Automatic · heard phrase with chime and Undo · running late with fixes
S09, S17, V2

Step list — CONTENT
Done · running · waiting
S11, S19, C1, C2

Ledger row — CONTENT
Latest with Undo · older with time
S14, S16

Call transcript — CONTENT
Maya · other party, hold timer, outcome card
S08

Info row — CONTENT
Icon tile, title, detail and status
S12, S20

Chip — CONTENT
Default · active · failed · state
S02, S03, S08, S14, S16

Badge — CONTENT
New · changed · cheapest · booked · done · limit
S03, S09, S17, S19, C4, V2

Button — INPUTS
Primary · secondary · text · destructive · call
All screens

Segmented control — INPUTS
Two or three options
S03, C5

Toggle — INPUTS
On · off
S03

Text field — INPUTS
Search · free text with caret · with Send
S05, S06

Keyboard — INPUTS
Letter · modifier · space · action key
S05

Form row — INPUTS
Label and value · with hint · selected
C5

Option row — INPUTS
Recommended · default, each with an action
V2

Incoming call — OVERLAYS
Answer · decline
S16

Intent card — OVERLAYS
Action, detail, progress and what comes next
S13

Takeover alert — OVERLAYS
Countdown ring, red, Level 3/4 variant only
V1

Sign-in — OVERLAYS
Face scan · PIN or password · switch profile
S01

Processing — OVERLAYS
"Building the plan" with sonar rings
S07

Arrival report — OVERLAYS
Large time, pill, what was done, next step
S20

Car top view — OVERLAYS
Default · selected compartment · door open · cabin scan
C1, C2, C5

Phone run — PHONE AND PANORAMIC VISION
App bar, live map card, steps: done · running · waiting
C2, C3

Camera still — PHONE AND PANORAMIC VISION
Detection frame, camera label, result chip
C3

Panoramic Vision band — PHONE AND PANORAMIC VISION
Presence · next stop · spare time · progress · event · horizon line in cyan, green, amber, red · offline · off
All Panoramic Vision states

Panoramic Vision

One band, always dark and not touchable, that shows nothing unique. Never on it: transcripts, maps, editing, anything long or interactive, medication names, personal data.



1 Automotive half
Speed, cabin temperature and range. Always on the left, always the same.

2 Presence
A small dot: listening, hearing you, speaking (rings), thinking (hollow), on a call (phone).

3 Next stop
Name and planned time. Replaced by an event for 4 seconds.

4 Spare time
The figure and the line share one colour: green, amber or red.

5 Progress
One square per stop: grey done, white current, dark to come.

6 Horizon line
Cyan under the car data, the spare-time colour under Maya. Flashes white on events.

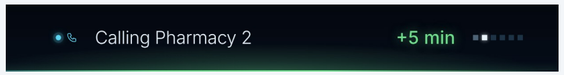
STATES THROUGH THE STORY · MAYA'S HALF

S02 Signed in



No plan yet: neutral line, no spare time.

S08 Pharmacy call



On a call. The plan exists, so spare time appears.

S09 Event · plan changed



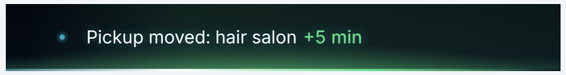
A change in the world: no chime, the line flashes white for 4 s.

S12 Plan running



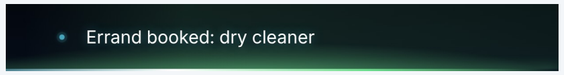
Next stop, spare time, one square per stop.

S17 Event · ambient change



Heard in the call: the chime sounds once, the line flashes.

S19 Event · errand booked



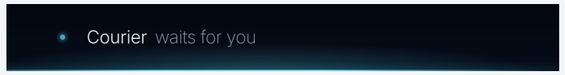
Agreed with the shop. Spare time unchanged.

S20 Arrival



Every square filled; the final spare time.

C1 Courier hand-off



What happens next: the car waits for Michael's call.

C2 Off · cabin empty



Nobody inside: the whole band is off.

Amber Low spare time · 0 to 4 min



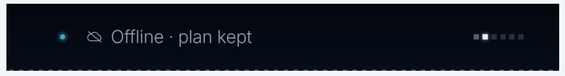
The number is always shown; the line turns amber.

V2 Red · below 0



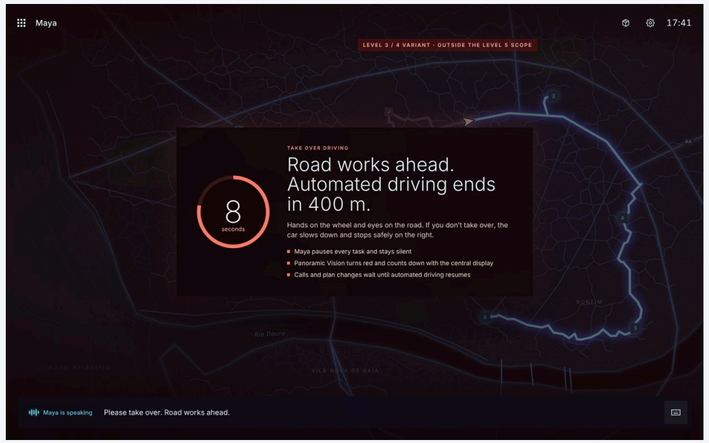
A minus sign and a warning mark, never colour alone.

Edge 8 Offline · plan kept



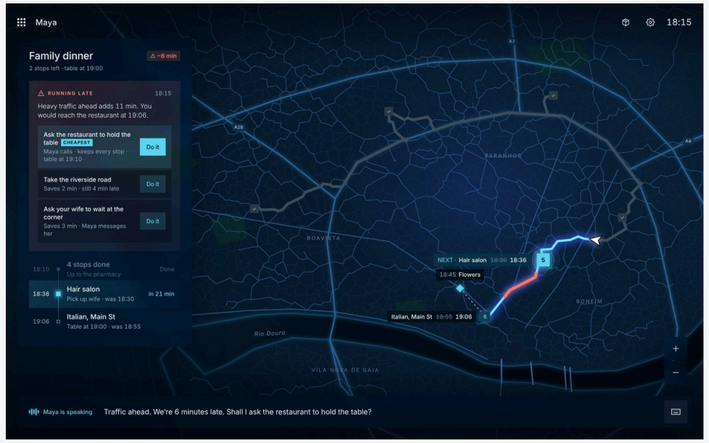
A grey dotted line; the plan keeps running.

Seventeen ways the evening could go wrong.



V1 Takeover request

Level 3 or 4 only, outside the Level 5 scope: an 8-second countdown on the central display and Panoramic Vision; Maya pauses every task and stays silent.



V2 Running late · edge case 2

−6 min with a warning mark. Three fixes, cheapest first, each with Do it. Maya speaks once, because the promise is at risk.

#	EDGE CASE	EXPECTED BEHAVIOUR
1	No pharmacy on the route has stock	Widen the search, show it on screen; ask only if the detour breaks spare time
2	Spare time goes negative	Fixes, cheapest first, such as holding the table (V2); Panoramic Vision turns red
3	Ambient false positive: a child says “let’s go to the park”	Act only when confident, otherwise ask; chime, heard phrase and Undo
4	The chime is missed	The change stays on screen until dismissed, and in the ledger; Panoramic Vision repeats it once
5	The florist can’t deliver before 19:00	Try the next florist; otherwise offer a pickup stop
6	A pharmacist refuses an AI caller	Next pharmacy, with the refusal logged
7	A child is not at the pickup point	Alert on the central display, a wait window, a call to the school
8	Connectivity loss	Keep the current plan; Panoramic Vision shows Offline; edits queue
9	Dinner time or place changes by message	Same pattern as the wife’s call: chime and Undo
10	Several people speak	Commands only from the signed-in user; other speech is context
11	Shop not ready, or nobody comes to the car	Wait within a set window, then leave; the report says why, and the errand returns to Errands
12	Wrong item loaded, or more than one	The camera checks before the door locks; Maya texts the shop and waits within the window
13	Someone or something left in the cabin	A person or a pet stops courier mode; a bag gets “Please take your belongings” and a note on the phone
14	The car is needed before the errand is done	Michael calls it; the same car comes from charging, or finishes the handover first
15	The errand runs late (traffic, a slow handover)	Tell Michael early on the phone; if he calls the car, it shows when it can reach him
16	Wrong door, or the code is shared	The code opens only the agreed door, once, at the agreed place and time
17	Item not allowed: too big, an animal, needs a signature	Declined while booking, before anything is promised

Cases 11 to 17 cover courier mode. Cases 1, 3 and 4 also appear in the prototype.

Riskiest first: does ambient intent feel helpful, or creepy? Would people send an empty car?

TESTS

TEST	METHOD	SUCCESS SIGNAL	CHECKS
Ambient-intent false positives	Scripted audio clips, true changes and decoys; measure Undo and reaction	Few false positives; chime and phrase understood	A3, A9
Courier trust test	Clickable courier flow, S18 to C3, 5 participants: what would they send, what would stop them?	4 of 5 would send it for an everyday errand	A14
Glance test	5-second exposure: “What’s next? How much spare time?”	80% or more correct	A8, A10
Replan legibility	Show a change, ask “what changed?”	Understood in under 5 s	A9
Touch-only run	Brief and edit a plan without speaking	Same completion as voice	A11
Wizard of Oz briefing	A person plays Maya’s voice, 5 participants (if I had a month)	No confusion; count the questions	A1, A6

The glance, legibility, touch-only and courier trust tests can run remotely on the clickable prototype. Scripted calls to real shops (A4, A13) and the Wizard of Oz session are on the one-month list.

ASSUMPTIONS · HIGH RISK IN RED

A1 Users accept “do first, show clearly, undo easily”

A2 Maya can predict the destination and learn the day from habit and data

A3 Listening to cabin talk and calls is welcome; false positives are cheap

A4 Retailers cooperate with a self-identified AI caller

A5 Automatic payments from Maya Wallet, up to a limit, are acceptable

A6 Users accept as many questions as needed, if each is necessary

A7 A suggestion tied to a known occasion is welcome

A8 Spare time gives at-a-glance confidence

A9 Chime, heard phrase and Undo are understood in a noisy cabin

A10 A passive Panoramic Vision carries enough at a glance

A11 Voice and touch parity is achievable for every task

A12 On-time arrival is the main success criterion

A13 Shops will hand items to an empty car that opens one door with a code

A14 People will send their empty car, and accept a cabin camera

A15 An empty Level 5 car may legally run errands

A16 Michael’s phone data (messages, calendar) is available to the car, despite platform limits

SUCCESS METRICS · ILLUSTRATIVE TARGETS

On-time arrival, within the 5 minutes of spare time	**95% or more**
Replans that keep spare time at or above zero	**90% or more**
Questions Maya could have answered herself	**Under 10%**
Undo rate on automatic and ambient changes	**Under 10%**
Time from a change to a visible adapted plan	**3 s or less**
Proactive suggestions accepted	**50% or more**
“Feels like a great assistant”	**4 / 5 or more**
Courier errands completed without Michael stepping in	**90% or more**
After a courier run, the car arrives within the time shown when called	**95% or more**
Actions moved from Ask first to Act within a month	**Rising**

What each choice gains, what it costs, and what I would do with a month.

DECISION	GAIN	COST
Level 5 only	One calm, trip-level experience	No driver-state design; V1 sketches Level 3 or 4
Full trust: Maya acts, pays and listens	Feels like an executive assistant	No approval before calls, or payments under the limit; trust lives in S03 and Undo
Maya speaks rarely	A quiet cabin	Voice-only users get less status; captions and Panoramic Vision help
Ambient intent: act when sure, ask when unsure	Plans follow life without being managed	False positives, answered by chime, heard phrase and Undo
Silent retailer calls, shown as transcripts	Silence in the cabin	Michael can read what Maya promised, not hear it
Banners stay until dismissed	Nothing is lost	Clutter if changes pile up
Undo on the latest ledger row only	A calm ledger, one clear action	Older actions are slower to reverse
Edit plan with a ripple preview, then Apply	Informed edits	An extra step for a quick change
A passive Panoramic Vision with nothing unique	No second interactive surface	It can never be the only channel
One layout: plan left, flat map	Learnable and consistent	No right-to-left variant, no 3D context
Courier: no handoff, no commitment	Errands don't fail at the door	Fewer errands qualify
Courier: one door, one code, a cabin camera	The rest of the car stays closed	A camera in the cabin; the item must fit the compartment
A solo exercise, no user research	Speed	A1 to A16 are untested

IF I HAD A MONTH

- Wizard of Oz sessions and a study of real family trips
- Scripted calls to real shops, for AI calls and empty-car handovers
- Other errand types beyond pick-ups and drop-offs
- Responsive layouts: each screen size designed, not scaled
- Trust that grows over time, a full activity log, wallet top-ups and statements
- Multi-passenger conflicts and escalation to a human
- Accessibility research with touch-only and voice-only users
- A regulation review for empty cars (A15)
- When Maya is unsure about a payment: ask first, or act and show Undo (leaning towards Undo, below a limit)
- Screens for later: the app launcher's contents, creating a new profile, idle or standby, guest mode
- A shared digital key for handovers, instead of a typed code
- The 3D head-up display
- Driving preferences, such as "prioritise speed over cost"

Thank you :)

“This is definitely the most fun challenge I’ve done for a job application. If I had no deadline, I wouldn’t stop pushing forward.”

Aníbal Oliveira · Senior Product Designer

Don’t forget to visit Maya’s official page:

oliveiraanibal.github.io/maya

The full UX artifacts.